Digital Notes

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Department of CSE (Data Science)

School of Engineering

UNIT-I: Sets and Relations

Syllabus

Sets, Operations on Sets, Venn Diagrams, Multi Sets, Binary Relations, Equivalence Relations, Ordering Relations, Operations on Relations, Partial Orders, Functions: Definition and Introduction, Composition of Functions, Inverse Functions, Binary and n-ary Operations

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Sets

Definition: "A set is a well-defined collection of objects." These objects are called elements or members of the set.

So, we can say that it is the collection of all such type of elements or objects which satisfy some rule and it is possible to say whether a particular object belongs to the collection or not.

We write $a \in A$ to denote that " a is an element of the set A". Similarly, $a \notin A$ is used to denote that " a is not an element of the set A". It is common for sets to be denoted by using uppercase letters. Lowercase letters are usually used to denote elements of sets.

Examples: 1. The set V of all vowels in the English alphabet can be written as $V = \{a, e, i, o, u\}$.

2. The set O of odd positive integers less than 10 can be expressed by $O = \{1,3,5,7,9\}$.

3. Although sets are usually used to group together elements with common properties, there is nothing that prevents a set from having seemingly unrelated elements. For instance, $\{a, 2, \text{Fred}, \text{New Jersey}\}$ is the set containing the four elements a, 2, Fred, and New Jersey.

4. The set of positive integers less than 100 can be denoted by $\{1, 2, 3, \ldots, 99\}$.

Remark: Sets can have other sets as members. For example, the set {**N**, **Z**,**Q**, **R**} is a set containing four elements, each of which is a set. The four elements of this set are **N**, the set of natural numbers; **Z**, the set of integers; **Q**, the set of rational numbers; and **R**, the set of real numbers.

Remark: Note that the concept of a datatype, or type, in computer science is built upon the concept of a set. In particular, a **datatype** or **type** is the name of a set, together with a set of operations that can be performed on objects from that set. For example, *boolean* is the name of the set $\{0,1\}$ together with operators on one or more elements of this set, such as AND, OR, and NOT.

Notations: There are five ways used to describe a set.

1. Describe a set by describing the properties of the members of the set **(Set builder notation)**.
2. Describe a set by listing its elements **(Roster method).**
3. Describe a set A by its characteristic function, defined as

$$\mu_A(x) = \begin{cases} 1 & \text{if} \quad x \in A \\ 0 & \text{if} \quad x \notin A \end{cases}$$

for all x in U where U is the universal set, sometimes called the "universe of discourse" or just the "universe" which is a fixed specified set describing the context for the duration of the discussion.

4. Describe a set by a recursive formula. This is to give one or more elements of the set and a rule by which the rest of the elements of the set may be generated.
5. Describe a set by an operation (such as union, intersection, complement, etc.) on some other sets.

Problem: Describe the set containing all the nonnegative integers less than or equal to 5.
Solution: Let A denote the set. Then the set A can be described in the following ways:

1. A = {x | x is a nonnegative integer less than or equal to 5}.
2. A = {0,1,2,3,4,5}.
3. $\mu_A(x) = \begin{cases} 1 & \text{for } x = 0,1,2,3,4,5 \\ 0 & \text{otherwise} \end{cases}$
4. $A = \{x_{i+1} = x_i + 1,\ i = 0,1,2,3,4,\ \text{where } x_0 = 0\}$.

Remark: Sometimes in roster method, a set is described without listing all its members. Some members of the set are listed, and then ***ellipses*** **(. . .)** are used when the general pattern of the elements is obvious.

Subset: Let A and B be two sets. Then A is said to be a subset of B if every element of A is an element of B.

If A is a subset of B, we say A is contained in B. Symbolically, we write $A \subseteq B$.

Proper Subset: A is said to be a proper subset of B if A is a subset of B and there is at least one element of B which is not in A.

If A is a proper subset of B, then we say A is strictly contained in B. Symbolically, we write $A \subset B$.

Properties: Let A, B, and C are sets. Then we have
1. $A \subseteq A$.
2. If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.
3. If $A \subseteq B$ and $B \subset C$ then $A \subset C$.
4. If $A \subseteq B$ and $A \not\subset C$ then $B \not\subset C$.

Equal Sets: Two sets A and B are equal iff $A \subseteq B$ and $B \subseteq A$. We write $A = B$.

Result: To show that two sets A and B are equal, we must show that each element of A is also an element of B, and conversely.

Null Set or Empty Set: A set containing no elements is called the empty set or null set, denoted by ϕ.
For example, given the universal set U of all positive numbers, the set of all positive numbers x in U satisfying the equation $x + 1 = 0$ is an empty set since there are no positive numbers which can satisfy this equation.

Note: The empty set is a subset of every set.

Note: It is important to note that the sets ϕ and $\{\phi\}$ are very different sets. The former has no elements, whereas the latter has the unique element ϕ.

Singleton: A set containing a single element is called a singleton.

Operations on Sets

There are several operations on set. A few important ones are discussed in this section. Consider U as the universal set and now we define the following operations.

Absolute Complement: Let A be any subset of U. The absolute complement of A is defined as $\bar{A} = \{x : x \notin A\}$.

Relative Complement: If A and B are sets, the relative complement of A with respect to B is given as $B - A = \{x : x \in B \text{ and } x \notin A\}$.

Note: $\bar{\phi} = U$, $\bar{U} = \phi$ and the complement of the complement of A is equal to A i.e. $\bar{\bar{A}} = A$.

Union: Let A and B be two sets. The union of A and B is given as

$$A \cup B = \{x : x \in A \text{ or } x \in B \text{ or both}\} .$$

More generally, if $A_1, A_2, ..., A_n$ are sets, then their union is the set of all objects which belong to at least one of them, and is denoted by

$$A_1 \cup A_2 \cup ... \cup A_n \text{ or } \bigcup_{j=1}^{n} A_j .$$

Intersection: Let A and B be two sets. The intersection of A and B is given as

$$A \cap B = \{x : x \in A \text{ and } x \in B\} .$$

More generally, if $A_1, A_2, ..., A_n$ are sets, then their intersection is the set of all objects which belong to every one of them, and is denoted by

$$A_1 \cap A_2 \cap ... \cap A_n \text{ or } \bigcap_{j=1}^{n} A_j .$$

Basic Properties	**Union**	**Intersection**
Idempotent	$A \cup A = A$	$A \cap A = A$
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$A \cup (B \cup C) = (A \cup B) \cup C$	$A \cap (B \cap C) = (A \cap B) \cap C$

Remark: It should be noted that, *in general*,

$$(A \cup B) \cap C \neq A \cup (B \cap C).$$

Symmetrical Difference or Boolean Sum: The symmetrical difference of two sets A and B is given as

$$A \Delta B = \{x : x \in A, \text{or } x \in B, \text{but not both}\} .$$

Disjoint Sets: Two sets A and B are said to be disjoint if they do not have a member in common.

Mathematically, $A \cap B = \phi$.

Distributive Laws: Let *A*, *B* and *C* are three sets. Then,

$$C \cap (A \cup B) = (C \cap A) \cup (C \cap B),$$

$$C \cup (A \cap B) = (C \cup A) \cap (C \cup B).$$

Power Set: Let *A* be a given set. The power set of *A*, denoted by *P(A)*, is the family of sets such that $X \subseteq A$ iff $X \in P(A)$.

Symbolically,

$$P(A) = \{X : X \subseteq A\}.$$

Example: Let $A = \{a, b, c\}$. Then the power set of *A* is

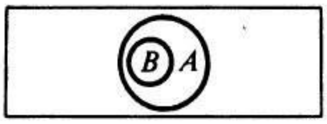
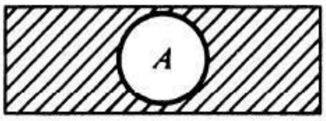
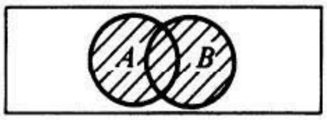
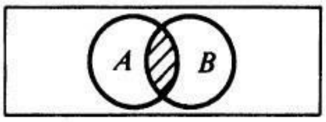
$$P(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{a, b, c\}\}.$$

Venn Diagram

It is often helpful to use Venn diagram [after JohnVenn (1834-1883)], to visualize the various properties of the set operations. The main points of Venn Diagram are:

- The universal set is represented by a large rectangular area.
- Subsets within this universe are represented by circular areas.

A summary of set operations and their Venn diagrams is given below:

Set Operation	Symbol	Venn Diagram
Set *B* is contained in set *A*	$B \subset A$	
The absolute complement of set *A*	$\bar{A}$	
The relative complement of set *B* with respect to set *A*	$A - B$	
The union of sets *A* and *B*	$A \cup B$	
The intersection of sets *A* and *B*	$A \cap B$	
The symmetrical difference of sets *A* and *B*	$A \Delta B$	

De Morgan's Laws: Let A and B be two sets. Then,

$$\overline{(A \cup B)} = \overline{A} \cap \overline{B}$$
$$\overline{(A \cap B)} = \overline{A} \cup \overline{B}$$

Note: The proof of the above laws can be given with the help of Venn diagram.

Problem: Prove that empty set is unique.

Proof: Let there are two empty sets, ϕ_1 and ϕ_2.

Since ϕ_1 and ϕ_2 are included in every set.

Therefore, $\phi_1 \subseteq \phi_2$ and $\phi_2 \subseteq \phi_1$

$\Rightarrow$ $\phi_1 = \phi_2$. Hence Proved.

Problem: Let $U = \{1,2,3,4,5\}$, $A = \{1,5\}$, $B = \{1,2,3,4\}$. Find the following sets.

(a) $\underline{A} \cap \underline{\overline{B}}$
(b) $A \cup B$

Solution: (a) $A \cap \overline{B} = \{1,5\} \cap (U - B)$

$= \{1,5\} \cap \{5\}$

$= \{5\}$

(b) $\overline{A} \cup \overline{B} = (U - A) \cup (U - B)$

$= \{2, 3, 4\} \cup \{5\}$

$= \{2, 3, 4, 5\}$

Multiset

In mathematics, a multiset (or bag, or mset) is a modification of the concept of a set that, unlike a set, allows for multiple instances for each of its elements.

The number of instances given for each element is called the **multiplicity** of that element in the multiset.

As a consequence, an infinite number of multisets exist which contain only elements *a* and *b*, but vary in the multiplicities of their elements:

- The set {*a*, *b*} contains only elements *a* and *b*, each having multiplicity 1 when {*a*, *b*} is seen as a multiset.
- In the multiset {*a*, *a*, *b*}, the element *a* has multiplicity 2, and *b* has multiplicity 1.
- In the multiset {*a*, *a*, *a*, *b*, *b*, *b*}, *a* and *b* both have multiplicity 3.

These objects are all different, when viewed as multisets, although they are the same set, since they all consist of the same elements. As with sets, and in contrast to tuples, order does not matter in discriminating multisets, so {*a*, *a*, *b*} and {*a*, *b*, *a*} denote the same multiset.

Note: To distinguish between sets and multisets, a notation that incorporates square brackets is sometimes used: the multiset {*a*, *a*, *b*} can be denoted as [*a*, *a*, *b*].

Remark: The cardinality of a multiset is constructed by summing up the multiplicities of all its elements. For example, in the multiset {*a*, *a*, *b*, *b*, *b*, *c*} the multiplicities of the members *a*, *b*, and *c* are respectively 2, 3, and 1, and therefore the cardinality of this multiset is 6.

Example: One of the simplest and most natural examples is the multiset of prime factors of a natural number *n*. Here the underlying set of elements is the set of prime factors of *n*. We know that $120 = 2^3\, 3^1\, 5^1$ which gives the multiset {2, 2, 2, 3, 5}.

Notation: We can write $a \in^n A$ if element *a* occurs in the multiset *A* at least *n* times.
For example: $1 \in^3 \{1,2,3,1,1\}$, $4 \in^0 \{1,2,3,1,1\}$.

Applications: Multiset Concept is useful in following areas:

- fundamental in combinatorics
- an important tool in the theory of relational databases
- used to implement relations in database systems
- SQL operates on multisets and return identical records

Relations

Ordered Pair: In ordered pair, each set is specified by two objects in a prescribed order. The ordered pair of *a* and *b*, with first coordinate *a* and second coordinate *b*, is the set (a,b).

Note: $(a,b) = (c,d)$ iff $a = c$ and $b = d$.

Cartesian Product: Let *A* and *B* are two sets. The Cartesian product of *A* and *B* is defined as

$$A \times B = \{(a,b) : a \in A \text{ and } b \in B\}.$$

More generally, the Cartesian product of *n* sets $A_1, A_2, ..., A_n$ is defined as

$$A_1 \times A_2 \times ... \times A_n = \{(a_1, a_2, ..., a_n) : a_i \in A_i, i = 1, 2, ..., n\}.$$

The expression $(a_1, a_2, ..., a_n)$ is called an ordered n-tuple.

Example: Let $A = \{0,1,2\}$ and $B = \{a,b\}$ Then,

$$A \times B = \{(0,a), (0,b), (1,a), (1,b), (2,a), (2,b)\}.$$

Note: From the definition of the Cartesian product we have seen that any element (a,b) in a Cartesian product $A \times B$ is just an ordered pair.

Binary Relation: A binary relation R from A to B is a subset of Cartesian product $A \times B$.

***n-a* ry Relation:** An n-ary relation is a subset of a Cartesian product of n sets $A_1, A_2, ..., A_n$.
In case $n = 1$, a subset R of A is called a **unary** relation on A.

Example: Let all the points inside a unit circle whose centre is at the origin. Then the set

$$R = \left\{ (x, y) : x \text{ and } y \text{ are real numbers and } x^2 + y^2 < 1 \right\}$$

is a relation on the set of real numbers.

Domain of R: Let R be a relation from A to B. The **domain** of R denoted by **dom *R***, is defined as

$$\text{dom } R = \{ x : x \in A \text{ and } (x, y) \in R \text{ for some } y \in B \}.$$

Range of R: Let R be a relation from A to B. The **range** of R denoted by **ran *R***, is defined as

$$\text{ran } R = \{ y : y \in B \text{ and } (x, y) \in R \text{ for some } x \in A \}.$$

Note: 1- Here, $\text{dom } R \subseteq A$ and $\text{ran } R \subseteq B$. Moreover, the domain of R is the set of first coordinates in R and the range of R is the set of second coordinates in R.

Note: 2- We sometimes write $(x, y) \in R$ as $x R y$ which reads "x relates to y".

Properties of Binary Relations

1	**Transitivity**	$\forall x, y, z$ if $x R y$ and $y R z$, then $x R z$;
2	**Reflexivity**	$\forall x$ $x R x$;
3	**Irreflexivity**	$\forall x$ $x \not R x$;
4	**Symmetry**	$\forall x, y$ if $x R y$, then $y R x$;
5	**Antisymmetry**	$\forall x, y$ if $x R y$ and $y R x$, then $x = y$;
6	**Asymmetry**	$\forall x, y$ if $x R y$, then $y \not R x$;

Equivalence Relation

Definition: Let R be a relation on A. R will be an equivalence relation on A if the following conditions are satisfied:

1. If xRx $\forall x \in A$ (R is reflexive).
2. If xRy, then yRx $\forall x, y \in A$ (R is symmetric).
3. If xRy and yRz, then xRz $\forall x, y, z \in A$ (R is transitive).

Example: Let $N = \{1, 2, 3, \ldots\}$ be the set of natural numbers. Define a relation R in **N** as follows:

$$R = \{ (x, y); x, y \in N \text{ and } x + y \text{ is even} \}.$$

R is an equivalence relation in N because the first two conditions are clearly satisfied and to prove the third condition, if $x+y$ and $y+z$ are divisible by 2, then $x+\left(y+y\right)+z$ is divisible by 2. Therefore, $x+z$ is divisible by 2. Hence Proved.

Note: In this equivalence relation all the odd numbers are equivalent and so are all the even numbers.

Partition: Let A be a given set. A partition of A is a collection P of disjoint subsets whose union is A.
Mathematically,

1. For any $B \in P$, $B \subseteq A$;
2. For any $B, C \in P$, $B \cap C = \phi$, or $B = C$; and
3. For any $x \in A$, $\exists\ B \in P$ such that $x \in B$.

Equivalence Class: Let A be a given set and R is an equivalence relation on A, the equivalence class $[x]$ of each element x of A is defined as

$$[x] = \{y \in A : xRy\}.$$

Note: We can have $[x]=[y]$ even if $x \neq y$ provided xRy.

Remark: Let A be a given set and R is an equivalence relation on A, then $S=\{[x] : x \in A\}$ is a partition of A into disjoint nonempty subsets. Conversely, if P is a partition of A into nonempty disjoint subsets, then P is the set of equivalence classes for the equivalence relation E defined on A by aEb iff a and b belong to the same subset of P.

Congruence modulo 'm': Let *'m'* be any positive integer then the relation congruence modulo *'m'* $\lfloor \lceil \equiv (\bmod\ m) \rceil \rfloor$ is defined by $x \equiv y \pmod{m}$ iff $x = y + a.m$ for some integer a.

Theorem: For any positive integer m, the relation $\equiv (\bmod\ m)$ is an equivalence relation on the integers, and partitions the integers into „m" distinct equivalence classes: [0], [1],…,[m-1].

Proof: We have $a \equiv b \pmod{m}$ which implies $a-b=m.k$.

First, we prove R is an equivalence relation. For this,

1. Reflexivity: $a-a=m.0$

$\Rightarrow \quad a \equiv a \pmod{m}$

$\Rightarrow \quad aRa$.

2. Symmetry: If $a \equiv b \pmod{m}$ then $a-b=m.k$

$\Rightarrow \quad -(a-b) = -m.k$

$\Rightarrow \quad b-a = m.(-k) = m.k'$ where $k'=k$

$\Rightarrow \quad b \equiv a \pmod{m}$

$\Rightarrow \qquad bRa.$

3. Transitivity: If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$

then $a - b = m.k_1$ and $b - c = m.k_2$

Adding these, we get

$\Rightarrow \qquad a-b+b-c = m.k_1 + m.k_2$

$\Rightarrow \qquad a-c = m.k_3$ where $k_3 = k_1 + k_2$

$\Rightarrow \qquad a \equiv c \pmod{m}.$

Therefore, it is an equivalence relation.

Now, if x is any integer, then by division algorithm

$$x = mq + r$$

where q and r are integers and $0 \le r < m$.

Thus, $\quad x \equiv r \pmod{m}$ and $[x] = [r]$.

So, each equivalence class for this relation is one of the classes [0],[1],…,[m-1].

Moreover, if $[x] = [y]$ where $0 \le x \le y \le m$-1,

Then $y = ma + x$ for some integer a.

Therefore, $\quad 0 \le y - x = m.a < m$ implies $a = 0$ and $x = y$.

Ordering Relations

In this topic, we need to understand the following concepts.

Order Theory: It studies various kinds of binary relations that capture the intuitive notion of ordering expressed usually using phrases "less than" or "precedes".

Partial Order: A relation R on a set A is called a partial order on A when R is reflexive, antisymmetric, and transitive, and then the set A is called a **partially ordered set** or a **poset**.

Notation: $[A; R]$ is used to denote that A is partially ordered by the relation R.

Note: The relation $\le$ on the set of real numbers is the prototype of a partial order, so it is common to write $\le$ to represent an arbitrary partial order on A.

Characteristic Properties of a Partial Order

1. Reflexivity i.e. $\forall\ a \in A,\ a \le a$

2. Antisymmetry i.e. $\forall\ a,b \in A$, if $a \le b$ and $b \le a$ then $a = b$
3. Transitivity i.e. $\forall\ a,b,c \in A$, if $a \le b$ and $b \le c$ then $a \le c$

Comparable: Two elements a and b in a set A are said to be **comparable** under $\le$ if either $a \le b$ or $b \le a$; otherwise they are **incomparable**.

Totally Ordered Set: If every pair of elements of a set A are comparable, then we say that $[A;\le]$ is **totally ordered** or that A is a **totally ordered set** or a **chain**.

In this case, the relation $\le$ is called a **total order**.

Examples: 1. If $\boldsymbol{Z}$ is the set of integers and $\le$ is the usual ordering on $\boldsymbol{Z}$, then $[Z;\le]$ is partially ordered and totally ordered.

2. The relation < on $\boldsymbol{Z}$ is not a partial order because it is not reflexive.

Functions

Definitions: 1. A **function** is a rule which maps a number or entity to another unique number or entity.

2. Let A and B be two non-empty sets. A **function** f from A to B is a relation from A to B such that:
(i) $dom\, f = A$ i.e. f is defined at each $a \in A$.
(ii) if $(x,y) \in f$ and $(x,z) \in f$ then $y = z$ i.e. no element of A is related to two elements of B.

Notes: 1. The words mapping, transformation, correspondence, and operator are used as synonyms of function.

2. If f is a function from A to B, we write $f : A \to B$.

One-One Function or Injective Function: A function $f : A \to B$ is said to be one-to-one if $f(x_1) = f(x_2) \quad \Rightarrow \quad x_1 = x_2$.

Thus, f is one-one iff for each $b \in \text{ran } f$, b has precisely one pre-image.

Onto Function or Surjective Function: A function $f : A \to B$ is said to be onto iff $\text{ran } f = B$.

In other words, f is onto if each $b \in B$ has some pre-image in A.

Bijective Function: If a function is both one-to-one and onto then it is called a bijective function.

Remark: A one-to-one, onto function $f: A \to B$ is usually called a one-to-one correspondence between *A* and *B*.

Note: 1. If a function $f: A \to B$ is not one-to-one, we call it a **many-to-one** function.

2. If $f: A \to B$ is not necessarily onto B, then it is said to be **into** B.

3. Since a function is a set, two functions f and g from A to B are equal if they are equal as sets. In other words, $f = g$ iff $f(a) = g(a)$ for each $a \in A$.

Constant Function: A function $f: X \to X$ is said to be a constant function if

$$f(x) = k \quad \forall\, x \in X \quad \text{where } k \text{ is fixed.}$$

Identity Function: A function $f: X \to X$ is said to be an identity function if

$$f(x) = x \quad \forall\, x \in X .$$

Example: Let $A = \{r, s, t\}$, $B = \{1, 2, 3\}$ and $C = \{r, s, t, u\}$.

Now, $R = \{(r,1), (r,2), (t,2)\}$ is a relation from *A* to *B* but *R* is not a function since image of *r* is not unique.

The set $f = \{(r,1), (s,2), (t,2)\}$ is a function from *A* to *B* but f is not one-to-one since 2 has two pre-images.. Likewise, f is not onto *B* since 3 has no pre-image.

The function $g = \{(r,1), (s,2), (t,3)\}$ is both a one-to-one and an onto function from *A* to *B*.

On the other hand, the function $h: C \to B$ defined as $h = \{(r,1), (s,1), (t,2), (u,3)\}$ is onto, but not one-to-one since 1 has two pre-images.

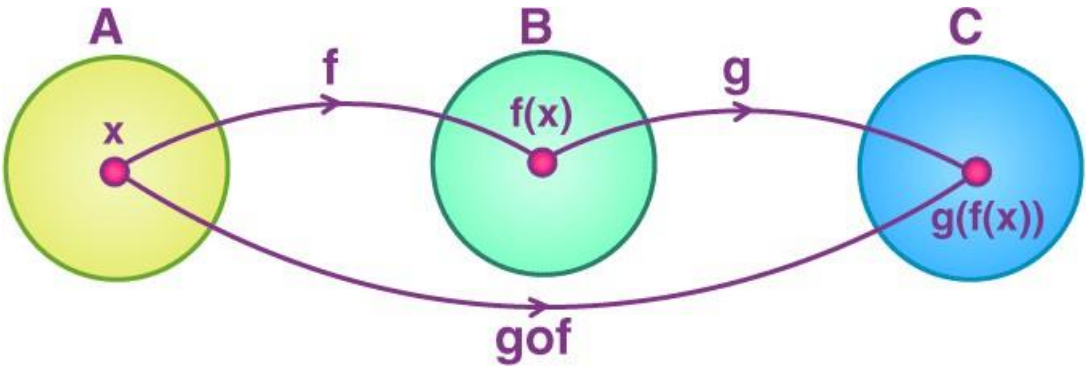
Composition of Functions

The composition of a function is an operation where two functions say f and g generate a new function say h in such a way that $h(x) = g(f(x))$.

It means here function g is applied to the function of x. So, basically, a function is applied to the result of another function.

Definition: Let $f: A \to B$ and $g: B \to C$ be two functions. Then the composition of f and g, denoted by $g \circ f$, is defined as the function $g \circ f: A \to C$ given by $g \circ f(x) = g(f(x))$, $\forall\, x \in A$.

The below figure shows the representation of **composite functions**.



Note: The order of function is an important thing while dealing with the composition of functions since $f \circ g(x)$ is not always equal to $g \circ f(x)$.

Symbol: It is also denoted as $g \circ f(x)$, where ∘ is a small circle symbol. We cannot replace o with a dot (.), because it will show as the product of two functions, such as $(g.f)(x)$.

Domain in Composition Function: $f(g(x))$ is read as f of g of x. In the composition of $f \circ g(x)$ the domain of function f becomes $g(x)$. The domain is a set of all values which go into the function.

Example: If $f(x)=3x+1$ and $g(x)=x^2$, then f of g of x, $f(g(x))=f(x^2)=3x^2+1$.

If we reverse the function operation, such as g of f of x, $g(f(x))=g(3x+1)=(3x+1)^2$.

Properties of Function Compositions

1. **Associative Property:** If there are three functions f, g and h, then they are said to be associative if and only if

$$f \circ (g \circ h) = (f \circ g) \circ h$$

2. **Commutative Property:** Two functions f and g are said to be commute with each other, if and only if

$$f \circ g = g \circ f$$

3. The function composition of one-to-one function is always one to one.
4. The function composition of two onto function is always onto.
5. The inverse of the composition of two functions f and g is equal to the composition of the inverse of both the functions, such as $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

Example: If $f\left(x\right)=3x^2$, then $f\circ f\left(x\right)=f\left(f\left(x\right)\right)=f\left(3x^2\right)=3.9x^4=27x^4.$

Inverse Function

Definition: An inverse function or an anti-function is defined as a function, which can reverse into another function. In simple words, if any function f takes x to y then, the inverse of f will take y to x.

If the function is denoted by f or F, then the inverse function is denoted by f^{-1} or F^{-1}.

One should **not confuse** (-1) with exponent or reciprocal here.

Note: If f and g are inverse functions, then $f\left(x\right)=y$ iff $g\left(y\right)=x.$

Example: If $f\left(x\right)=2x+5$ then what will be the inverse of f?

Let $f\left(x\right)=2x+5=y\Rightarrow x=\dfrac{y-5}{2}=g\left(y\right)$

Then, $f^{-1}\left(x\right)=\dfrac{x-5}{2}.$

Remark: In trigonometry, the inverse sine function is used to find the measure of angle for which sine function generated the value.

For example, $\sin^{-1}(1)=\sin^{-1}(\sin 90)=90$ degrees. Hence, sin 90 degrees is equal to 1.

Note: A function that consists of its inverse fetches the original value.

Types of Inverse Function: There are various types of inverse functions like the inverse of trigonometric functions, rational functions, hyperbolic functions and log functions. The inverses of some of the most common functions are given below.

Function	**Inverse of the Function**	**Comment**
+	–	
×	/	Don"t divide by 0
1/x	1/y	x and y not equal to 0
x^2	$\sqrt{y}$	x and $y \geq 0$
x^n	$y^{1/n}$	n is not equal to 0
e^x	ln(y)	y > 0
a^x	log a(y)	y and a > 0
Sin (x)	Sin^{-1} (y)	$-\pi/2$ to $+\pi/2$

Function	Inverse of the Function	Comment
Cos (x)	Cos^{-1} (y)	0 to π
Tan (x)	Tan^{-1} (y)	– π/2 to + π/2

Operations

We can classify the operations into following main categories:

Unary Operation: A unary operation is an operation with only one operand, i.e. a single input.

Examples: The function $f : A \to A$, (where A is a set) is a unary operation on A. The trigonometric functions are unary operations.

Notations: Common notations are prefix notation (e.g. +, −, ¬), postfix notation (e.g. factorial *n!*), functional notation (e.g. *sin x* or *sin(x)*), and superscripts (e.g. transpose A^T).

Binary Operation: A binary operation or dyadic operation is a calculation that combines two elements (called operands) to produce another element.

Examples: The familiar arithmetic operations of addition, subtraction, and multiplication. Other examples are readily found in different areas of mathematics, such as vector addition and matrix multiplication.

Remark: More precisely, a binary operation on a set S is a mapping of the elements of the Cartesian product $S \times S$ to S i.e. $f : S \times S \to S$.

Note: Because the result of performing the operation on a pair of elements of S is again an element of S, the operation is called a closed (or internal) binary operation on S (or sometimes expressed as having the property of closure).

n-ary Operation: An operation that takes n arguments for its input is an n-ary operation.

Example: Adding a bunch of numbers together would be a simple n-ary operation, i.e.

$$\text{Sum}(x_1, x_2, ..., x_n) = \sum_{i=1}^{n} x_i$$

Also, picking out a maximum, minimum, mean, medium, the first object in a list and so on are also its examples.

Remark: In general, a function takes an object from space X and maps it to space Y. If the objects in X are tuples (an ordered collection of sub-objects) then the function is an n-ary operation.

$$f : X \to Y$$

$$X = X_1 \times X_2 \times ... \times X_n$$

then f is an n-ary operation.

SOLVED PROBLEMS

Example 1. Which of these sets are equal: $\{x, y, z\}$, $\{z, y, z, x\}$, $\{y, x, y, z\}$, $\{y, z, x, y\}$?

Solution: They are all equal. Order and repetition do not change a set.

Example 2. List the elements of each set where $\mathbf{N} = \{1, 2, 3, \ldots\}$.

(a) $A = \{x \in \mathbf{N} \mid 3 < x < 9\}$

(b) $B = \{x \in \mathbf{N} \mid x \text{ is even}, x < 11\}$

(c) $C = \{x \in \mathbf{N} \mid 4 + x = 3\}$

Solution: (*a*) A consists of the positive integers between 3 and 9; hence $A = \{4, 5, 6, 7, 8\}$.

(b) B consists of the even positive integers less than 11; hence $B = \{2, 4, 6, 8, 10\}$.

(c) No positive integer satisfies $4 + x = 3$; hence $C = \emptyset$, the empty set.

Example 3. In a survey of 120 people, it was found that:

65 read *Newsweek* magazine, 20 read both *Newsweek* and *Time*,

45 read *Time*, 25 read both *Newsweek* and *Fortune*,

42 read *Fortune*, 15 read both *Time* and *Fortune*,

8 read all three magazines.

(a) Find the number of people who read at least one of the three magazines.

(b) Draw its Venn diagram.

(c) Find the number of people who read exactly one magazine.

Solution: (*a*) We want to find $n(N \cup T \cup F)$.

Using Inclusion–Exclusion Principle, we can write

$$n(N \cup T \cup F) = n(N) + n(T) + n(F) - n(N \cap T) - n(N \cap F) - n(T \cap F) + n(N \cap T \cap F)$$

$$= 65 + 45 + 42 - 20 - 25 - 15 + 8 = 100$$

(b) To draw the required Venn diagram, we need to obtain the following:

8 read all three magazines,

$20 - 8 = 12$ read *Newsweek* and *Time* but not all three magazines,

$25 - 8 = 17$ read *Newsweek* and *Fortune* but not all three magazines,

$15 - 8 = 7$ read *Time* and *Fortune* but not all three magazines,

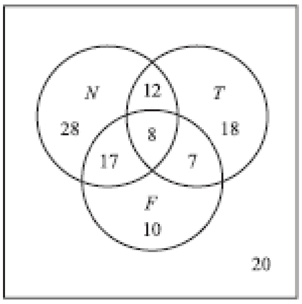
$65 - 12 - 8 - 17 = 28$ read only *Newsweek*,

$45 - 12 - 8 - 7 = 18$ read only *Time*,

$42 - 17 - 8 - 7 = 10$ read only *Fortune*,

$120 - 100 = 20$ read no magazine at all.

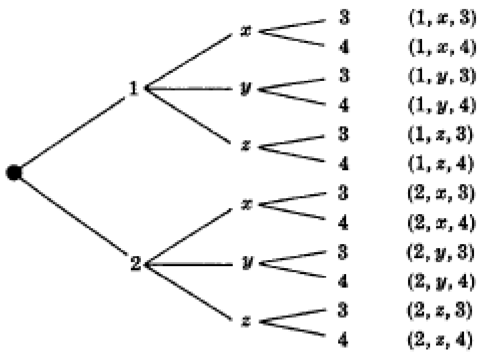
Hence, it can be drawn as following



(c) From Venn diagram, we can clearly see that 28+ 18 + 10 = 56 read exactly one of the magazines.

Example 4. Given: $A = \{1, 2\}, B = \{x, y, z\}$, and $C = \{3, 4\}$. Find: $A \times B \times C$.

Solution: $A \times B \times C$ consists of all ordered triplets (a, b, c) where $a \in A$, $b \in B$, $c \in C$. These elements of $A \times B \times C$ can be systematically obtained by a so-called tree diagram as shown below. The elements of $A \times B \times C$ are precisely the 12 ordered triplets to the right of the tree diagram.



Observe that $n(A) = 2$, $n(B) = 3$, and $n(C) = 2$ and, as expected,

$$n(A \times B \times C) = 12 = n(A) \cdot n(B) \cdot n(C).$$

Example 5. Find x and y given $(2x, x + y) = (6, 2)$.

Solution: Two ordered pairs are equal if and only if the corresponding components are equal. Hence we obtain the equations

$$2x = 6 \text{ and } x + y = 2$$

from which we derive the answers $x = 3$ and $y = -1$.

Example 6. If $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$. Let R be the following relation from A to B:

$$R = \{(1, y), (1, z), (3, y), (4, x), (4, z)\}$$

(a) Determine the matrix of the relation.

(b) Draw the arrow diagram of R.

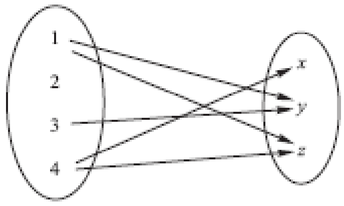
(c) Find the inverse relation R^{-1} of R.

(d) Determine the domain and range of R.

Solution:

$$\begin{array}{c|ccc} & x & y & z \\ \hline 1 & 0 & 1 & 1 \\ 2 & 0 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 4 & 1 & 0 & 1 \end{array}$$

(a)



(b)

(c) Reverse the ordered pairs of R to obtain R^{-1}:

$$R^{-1} = \{(y, 1), (z, 1), (y, 3), (x, 4), (z, 4)\}$$

Observe that by reversing the arrows in above figure, we obtain the arrow diagram of R^{-1}.

(d) The domain of R, Dom(R), consists of the first elements of the ordered pairs of R, and the range of R, Ran(R), consists of the second elements. Thus,

$$\text{Dom}(R) = \{1, 3, 4\} \text{ and } \text{Ran}(R) = \{x, y, z\}$$

Example 7. Let $X = \{1, 2, 3, 4\}$. Determine whether each relation on X is a function from X into X.

(a) $f = \{(2, 3), (1, 4), (2, 1), (3.2), (4, 4)\}$

(b) $g = \{(3, 1), (4, 2), (1, 1)\}$

(c) $h = \{(2, 1), (3, 4), (1, 4), (2, 1), (4, 4)\}$

Solution: Recall that a subset f of $X{\times}X$ is a function f: $X \to X$ if and only if each $a \in X$ appears as the first coordinate in exactly one ordered pair in f.

(a) No. Two different ordered pairs (2, 3) and (2, 1) in f have the same number 2 as their first coordinate.

(b) No. The element $2 \in X$ does not appear as the first coordinate in any ordered pair in g.

(c) Yes. Although $2 \in X$ appears as the first coordinate in two ordered pairs in h, these two ordered pairs are equal.

Example 8. Let $A = \{a, b, c\}$, $B = \{x, y, z\}$,$C = \{r, s, t\}$. Let f: $A \to B$ and g: $B \to C$ be defined by:

$$f = \{(a, y), (b, x), (c, y)\} \text{ and } g = \{(x, s), (y, t), (z, r)\}.$$

Find: (a) composition function $g{\circ}f$: $A \to C$; (b) Im(f), Im(g), Im$(g{\circ}f)$.

Solution: (a) Use the definition of the composition function to compute:

$$(g{\circ}f)(a) = g(f(a)) = g(y) = t$$

$$(g{\circ}f)(b) = g(f(b)) = g(x) = s$$

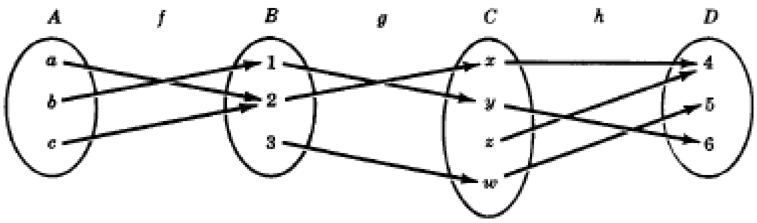
$$(g{\circ}f)(c) = g(f(c)) = g(y) = t$$

That is $g \circ f = \{(a, t), (b, s), (c, t)\}$.

(b) Find the image points (or second coordinates):

$$\mathrm{Im}(f) = \{x, y\},\ \mathrm{Im}(g) = \{r, s, t\},\ \mathrm{Im}(g \circ f) = \{s, t\}.$$

Example 9. Let the functions $f : A \to B$, $g : B \to C$, $h : C \to D$ be defined as in the following figure: Determine if each function is: (a) onto, (b) one-to-one, (c) invertible.



Solution: (a) The function $f : A \to B$ is not onto since $3 \in B$ is not the image of any element in A.

The function $g : B \to C$ is not onto since $z \in C$ is not the image of any element in B.

The function $h : C \to D$ is onto since each element in D is the image of some element of C.

(b) The function $f : A \to B$ is not one-to-one since a and c have the same image 2.

The function $g : B \to C$ is one-to-one since 1, 2 and 3 have distinct images.

The function $h : C \to D$ is not one-to-one since x and z have the same image 4.

(c) No function is one-to-one and onto; hence no function is invertible.

If you have any queries, comments or suggestions for the improvement of this note, please write to me at abhinava@mallareddyuniversity.ac.in . I shall be happy to update it.

Faculty and Course Coordinator -- Dr. M. Shravani

UNIT-II: Mathematical Logic and Induction

Syllabus

Statements and Notation, Connectives, Quantified Propositions, Logical Inferences, Methods of Proof of an Implication, First Order Logic and other Methods of Proof, Rules of Inference for Quantified Propositions, Proof by Mathematical Induction

Faculty and Course Coordinator – Dr. Abhinava Srivastav

1 Mathematical Logic : Statements and Notation

The rules of logic specify the meaning of mathematical statements. For instance, these rules help us understand and reason with statements such as "There exists an integer that is not the sum of two squares" and "For every positive integer n, the sum of the positive integers not exceeding n is $n(n + 1)/2$." Logic is the basis of all mathematical reasoning, and of all automated reasoning. It has practical applications to the design of computing machines, to the specification of systems, to artificial intelligence, to computer programming, to programming languages, and to other areas of computer science, as well as to many other fields of study.

Usually sentences are classified in to four types.

- Declarative
- Interrogative
- Exclamatory
- Imperative

In logics, we pay our attention to those declarative sentences to which it is meaningful to assign one and only one of the truth values TRUE or FALSE. Such declarative sentences are called *Propositions* or *Statements*. For definiteness the following assumptions are made about propositions.

Assumption 1 : *The Law of Excluded Middle*

For every proposition p, either p is true or p is false

Assumption 2 : *The Law of Contradiction*

For every proposition p, it is not the case that p is both true and false

2 Connectives

Propostions are combined by means of connectives to form new propositions or statements. They are

(i) not (ii) and (iii) or (iv) if .. then (v) If and only if

2.1 Negation (*not*)

If p is a Proposition, then *"p is not true"* is also a proposition. This we represent as "$\sim p$" and it is referred as "*not p*" or "negation of p" or "denial of p". The proposition " not p" is true when p is false and false when p is true.

Example : p : Ramanujan is genius

$\sim p$: It is not the case that Ramanujan is genius

or $\sim p$: It is false that Ramanujan is genius

or $\sim p$: Ramanujan is not genius

The truth table for negation is as given below.

p	$\sim p$
T	F
F	T

2.2 Conjunction (*and*)

If *p* and *q* are propositions, then *"p and q"* is also a proposition, which we represent as "$p \wedge q$" and is called "Conjunction". The conjunction "$p \wedge q$" is true only when *p* is true and *q* is true. In all other cases it is false.

Example : p : $\sqrt{2}$ is an irrational number
q : 7 is a prime number

$p \wedge q$: $\sqrt{2}$ is an irrational number and 7 is a prime number

The truth table for conjunction is as given below.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

2.3 Disjunction (*or*)

It is divided into two parts:

2.3.1 Inclusive Disjunction

If *p* and *q* are propositions, then *"p or q"* is also a proposition, represented by "$p \vee q$" and is called "Inclusive Disjunction". This disjunction "$p \vee q$" is true whenever at least one of the two propositions is true. In other words, this disjunction is false only when *p* is false and *q* is false. In all other cases, it is true.

Example : "Students who have taken calculus or computer science can take this class."

Here, we mean that students who have taken both calculus and computer science can take the class, as well as the students who have taken only one of the two subjects.

The truth table for inclusive disjunction is as given below.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

2.3.2 Exclusive Disjunction

If *p* and *q* are propositions, then *"p or q"* is also a proposition, represented by "$p \veebar q$" and is called "Exclusive Disjunction". This disjunction "$p \veebar q$" is true whenever one of the two propositions is true. In other words, this disjunction is false when *p* and *q* are both false or both true.

Example : "Students who have taken calculus or computer science, but not both, can enroll in this class." Here, we mean that students who have taken both calculus and computer science course cannot take the class. Only those who have taken exactly one of the two courses can take the class.

Similarly, when a menu at a restaurant states, "Soup or salad comes with an entrée," the restaurant almost always means that customers can have either soup or salad, but not both.

Hence, this is an exclusive, rather than an inclusive, or. The truth table for exclusive disjunction is given below.

p	q	$p \veebar q$
T	T	F
T	F	T
F	T	T
F	F	F

2.4 Conditional (*if* p *then* q)

If p and q are propositions, then the proposition *"p implies q"* or *"if p then q"* is represented as "$p \rightarrow q$" and is called an "Implication" or a "Conditional". In this case, p is called premise, hypothesis or antecedent of implication and q is called the conclusion or consequent of the implication. The conditional "$p \rightarrow q$" is false only when p is true and q is false, in all other cases it is true.

The statement "$p \rightarrow q$" is called a conditional statement because "$p \rightarrow q$" asserts that q is true on the condition that p holds. Note that the statement "$p \rightarrow q$" is true when both (i) p and q are true and (ii) p is false (no matter what truth value q has). Because conditional statements play such an essential role in mathematical reasoning, a variety of terminology is used to express "$p \rightarrow q$". In general, the following ways are used to express this conditional statement:

"if p, then q" "p implies q"
"if p, q" "q follows from p"
"p is sufficient for q" "a sufficient condition for q is p"
"q if p" "q whenever p"
"q when p"

Example: A useful way to understand the truth value of a conditional statement is to think of an obligation or a contract. For example, the pledge many politicians make when running for office is

"If I am elected, then I will lower taxes."

If the politician is elected, voters would expect this politician to lower taxes. Furthermore, if the politician is not elected, then voters will not have any expectation that this person will lower taxes, although the person may have sufficient influence to cause those in power to lower taxes. It is only when the politician is elected but does not lower taxes that voters can say that the politician has broken the campaign pledge. This last scenario corresponds to the case when p is true but q is false in "$p \rightarrow q$".

Similarly, consider a statement that a professor might make:

"If you get 100% on the final, then you will get an A."

If you manage to get a 100% on the final, then you would expect to receive an A. If you do not get 100% you may or may not receive an A depending on other factors. However, if you do get 100%, but the professor does not give you an A, you will feel cheated.

The truth table for conditional is as given below.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Converse, Contrapositive and Inverse: We can form some new conditional statements starting with a conditional statement "$p \to q$". In particular, there are three related conditional statements that occur so often that they have special names.

(i) The proposition "$q \to p$" is called the converse of "$p \to q$".

(ii) The contrapositive of "$p \to q$" is the proposition "$\neg q \to \neg p$".

(iii) The proposition "$\neg p \to \neg q$" is called the inverse of "$p \to q$".

Example: What are the contrapositive, the converse, and the inverse of the conditional statement "The home team wins whenever it is raining"?

Solution: Since "q whenever p" is one of the ways to express the conditional statement "$p \to q$", the original statement can be rewritten as

"If it is raining, then the home team wins."

So, we can write as following:

Contrapositive: "If the home team does not win, then it is not raining."

Converse: "If the home team wins, then it is raining."

Inverse: "If it is not raining, then the home team does not win."

Note: Out of these three conditional statements formed from "$p \to q$", only the contrapositive always has the same truth value as "$p \to q$".

2.5 Biconditional

If p and q are propositions, then the conjunction of conditionals $p \to q$ and $q \to p$ is called "Biconditional" of p and q and is denoted by $p \leftrightarrow q$. Thus $p \leftrightarrow q$ is same as $(p \to q) \wedge (q \to p)$ and is read as *"p if and only if q"*.

Example: "You can take the flight if and only if you buy a ticket."
This statement is true if p and q are either both true or both false, that is, if you buy a

ticket and can take the flight or if you do not buy a ticket and you cannot take the flight. It is false when p and q have opposite truth values, that is, when you do not buy a ticket, but you can take the flight (such as when you get a free trip) and when you buy a ticket but you cannot take the flight (such as when the airline bumps you).

The truth table for biconditional is as given below.

p	q	$p \to q$	$q \to p$	$p \leftrightarrow q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Combined Truth Table: The combined truth table is as follows:

p	q	$\sim p$	$p \wedge q$	$p \vee q$	$p \underline{\vee} q$	$p \to q$	$p \leftrightarrow q$
T	T	F	T	T	F	T	T
T	F	F	F	T	T	F	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

3 Definitions

Compound Propositions: The new propositions obtained by using connectives are called molecular or compound propositions.

Simple Propositions: Propositions which don't contain any logical connectives are called simple propositions.

Propositional Function: A propositional function is a function whose variables are propositions.

Remark: While representing a statement (proposition) involving connectives in symbolic form, care has to be taken to ensure that the symbolic representation conveys the intended meaning of the statement without any ambiguity. Appropriate parenthesis are to be used at appropriate places to achieve this objective.

Well-formed Formulae: Statements represented in symbolic forms which cannot be interpreted in more than one way are called well-formed formulae.

Logically Equivalent or Equivalent: Two propositional functions p and q are logically equivalent or equivalent, if they have same truth values or truth tables. Then we write $p \equiv q$.

Problem 1 Construct the truth table for the following:

(i) $p \wedge \sim q$ (ii) $\sim p \vee q$ (iii) $\sim p \veebar \sim q$ (iv) $p \rightarrow \sim q$

Solution :

p	q	$\sim p$	$\sim q$	$p \wedge \sim q$	$\sim p \vee q$	$\sim p \veebar \sim q$	$p \rightarrow \sim q$
T	T	F	F	F	T	F	F
T	F	F	T	T	F	T	T
F	T	T	F	F	T	T	T
F	F	T	T	F	T	F	T

Problem 2 Construct the truth tables of the following compound propositions:

(i) $(p \vee q) \wedge r$ (ii) $p \vee (q \wedge r)$

Solution :

p	q	r	$p \vee q$	$(p \vee q) \wedge r$	$q \wedge r$	$p \vee (q \wedge r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	T	T	F	T
T	F	F	T	F	F	T
F	T	T	T	T	T	T
F	T	F	T	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

Problem 3 Construct the truth tables of the following compound propositions:

(i) $(p \wedge q) \rightarrow r$ (ii) $q \wedge (\sim r \rightarrow p)$

Solution : Do it yourself.

Tautology: A compound proposition function which is true for all possible truth values of its components is called *"tautology"*.

Contradiction: A compound proposition function which is false for all possible truth values of its components is called *"contradiction"* or an *"absurdity"*

Contingency: A compund proposition which is neither a tautology nor a contradiction is called a *"contingency"*.

Example :

p	$\sim p$	$p \vee \sim p$	$p \wedge \sim p$
T	F	T	F
F	T	T	F

Here, $p \vee \sim p$ is a tautology, and $p \wedge \sim p$ is a contradiction.

Problem 4 Show that

(i) $(p \veebar q) \vee (p \leftrightarrow q)$ is a tautology,

(ii) $(p \veebar q) \wedge (p \leftrightarrow q)$ is a contradiction,

(iii) $(p \veebar q) \wedge (p \rightarrow q)$ is a contingency.

Solution :

p	q	$p \veebar q$	$p \rightarrow q$	$p \leftrightarrow q$	$(p \veebar q) \vee (p \leftrightarrow q)$	$(p \veebar q) \wedge (p \leftrightarrow q)$	$(p \veebar q) \wedge (p \rightarrow q)$
T	T	F	T	T	T	F	F
T	F	T	F	F	T	F	F
F	T	T	T	F	T	F	T
F	F	F	T	T	T	F	F

Problem 5 Prove that $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology.

Solution :

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

Hence, $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a Tautology.

Problem 6 For any two Propositions p, q prove that $(p \rightarrow q) \Leftrightarrow (\sim p \vee q)$.

Solution :

p	q	$p \rightarrow q$	$\sim p$	$\sim p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

We observe that $p \rightarrow q$ and $\sim p \vee q$ have identical truth values for all possible truth values of p and q.

$\therefore$ $(p \rightarrow q) \Leftrightarrow (\sim p \vee q)$ i.e., $p \rightarrow q$ and $\sim p \vee q$ are logically equivalent.

Logical Implication: A proposition p logically implies a proposition q, and q is a logical consequence of p, if the implication $(p \rightarrow q)$ is true for all possible assignments of the truth values of p and q, that is, if $(p \rightarrow q)$ is a tautology.

Note: Much care must be taken not to confuse implication (or conditional) with logical implication. The conditional is only a way of connecting the two propositions p and q, whereasif p logically implies q then p and q are related to the extent that whenever p

has the truth value T then so does q. We do note that **every logical implication is an implication (conditional), but not all implications are logical implications.**

Valid and Faulty Inferences: We say that an inference is **valid** if the implication is a tautology, that is, if the implication is a logical implication otherwise, we say that the inference is **faulty** or **invalid**.

4 The Laws of Logic

The following results are known as Laws of Logic obtained based on the definition of logical equivalence.

1. ***Law of double negation*** : For any Proposition p, $\sim(\sim p) \Leftrightarrow p$
2. ***Idempotent Laws*** : For any Proposition p, $(a)\ p \vee p \Leftrightarrow p$ $\quad (b)\ p \wedge p \Leftrightarrow p$
3. ***Identity Laws*** : For any Proposition p, $(a)\ p \vee F_0 \Leftrightarrow p$ $\quad (b)\ p \wedge T_0 \Leftrightarrow p$
   Where F_0 is a Contradiction and T_0 is a Tautology.
4. ***Inverse Laws*** : For any Proposition p, $(a)\ p \vee \sim p \Leftrightarrow T_0$ $\quad (b)\ p \wedge \sim p \Leftrightarrow F_0$
5. ***Domination Laws*** : For any Proposition p, $(a)\ p \vee T_0 \Leftrightarrow T_0$ $\quad (b)\ p \wedge F_0 \Leftrightarrow F_0$
6. ***Commutative Laws*** : For any two Proposition p and q,
   (a) $(p \vee q) \Leftrightarrow (q \vee p)$ $\quad (b)\ (p \wedge q) \Leftrightarrow (q \wedge p)$
7. ***Absorption Laws*** : For any two Proposition p and q,
   (a) $[p \vee (p \wedge q)] \Leftrightarrow p$ $\quad (b)\ [p \wedge (p \vee q)] \Leftrightarrow p$
8. ***De Morgan Laws*** : For any two Proposition p and q,
   $(a)\ \sim(p \vee q) \Leftrightarrow \sim p \wedge \sim q$ $\quad (b)\ \sim(p \wedge q) \Leftrightarrow \sim p \vee \sim q$
9. ***Associative Laws*** : For any three Proposition p, q and r
   (a) $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$ $\quad (b)\ p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$
10. ***Distributive Laws*** : For any three Proposition p, q and r
    (a) $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$ $\quad (b)\ p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

Problem 7 Obtain the truth table for $p \vee (q \wedge r)$ and $(p \wedge q) \vee (p \wedge r)$ for all possible truth values of p, q, r.

Solution : Do it yourself.

5 Fallacies

These are the statements which appear to be true but they are not. There are three forms of faulty inferences.

1. ***Fallacy of affirming the Consequent*** :

$$\begin{array}{l} p \rightarrow q \\ \underline{q \quad\quad} \\ \therefore \quad p \end{array} \quad \text{is a Fallacy}$$

since $[(p \rightarrow q) \wedge q] \rightarrow p$ is not a tautology.

2. ***Fallacy of denying the antecedent*** :

$$\begin{array}{l} p \rightarrow q \\ \underline{\sim p \quad\quad} \\ \therefore \quad \sim q \end{array} \quad \text{is a Fallacy}$$

since $[(p \rightarrow q) \wedge (\sim p)] \rightarrow (\sim q)$ is not a tautology.

3. ***Non sequitur fallacy*** (***Non sequitur means it doesn't follow***) :

$$\begin{array}{l} \underline{p \quad\quad} \\ \therefore \quad q \end{array} \quad \text{is a Fallacy}$$

since $p \rightarrow q$ is not a tautology.

Problem 8 Simplify the following Compound propositions using laws of logic.

(i) $(p \vee q) \wedge \sim(\sim p \vee q)$

(ii) $\sim [\sim [(p \vee q) \wedge r] \vee \sim q]$

Solution : The solution can be written as following:

(i) $(p \vee q) \wedge \sim(\sim p \vee q) \equiv (p \vee q) \wedge (p \wedge \sim q)$ DeMorgan and Double Negation Laws

$\equiv [(p \vee q) \wedge p] \wedge \sim q$ Associative Law

$\equiv [p \wedge (p \vee q)] \wedge \sim q$ Commutative Law

$\equiv p \wedge \sim q$ Absorption Law

(ii) $\sim [\sim [(p \vee q) \wedge r] \vee \sim q] \equiv \sim [\sim [(p \vee q) \wedge r]] \wedge \sim [\sim q]$ De Morgan Law

$\equiv [(p \vee q) \wedge r] \wedge q$ Law of double negation

$\equiv (p \vee q) \wedge (r \wedge q)$ Associative Law

$\equiv (p \vee q) \wedge (q \wedge r)$ Commutative Law

$\equiv ((p \vee q) \wedge q) \wedge r$ Associative Law

$\equiv q \wedge r$ Absorption Law

Remark : Logically equivalent propositions are treated as equivalent propositions. Some of the useful logically equivalent propositions are here under.

(i) $\sim (p \vee q) \equiv \sim p \wedge \sim q$ (ii) $\sim (p \wedge q) \equiv \sim p \vee \sim q$

(iii) $\sim (p \rightarrow q) \equiv p \wedge \sim q$

(iv) $(p \rightarrow q) \equiv \sim (\sim(p \rightarrow q)) \equiv \sim (p \wedge \sim q) \equiv \sim p \vee \sim\sim q \equiv \sim p \vee q$

Proposition	Negation
$\sim p$	p
$p \wedge q$	$\sim p \vee \sim q$
$p \vee q$	$\sim p \wedge \sim q$
$p \rightarrow q$	$p \wedge \sim q$
$\sim (p \rightarrow q)$	$\sim p \vee q$

Problem 9 Show that $(p \rightarrow q) \wedge [\sim q \wedge (r \vee \sim q)] \Leftrightarrow \sim(q \vee p)$.

Solution : This can be done by using truth tables of both LHS and RHS or obtain RHS from LHS by simplifying LHS.

$(p \rightarrow q) \wedge [\sim q \wedge (r \vee \sim q)] \equiv (p \rightarrow q) \wedge [\sim q \wedge (\sim q \vee r)]$ Commutative Law

$\equiv (p \rightarrow q) \wedge \sim q$ Absorption Law

$\equiv \sim [(p \rightarrow q) \rightarrow q]$ $\{\because \sim (p \rightarrow q) \equiv p \wedge \sim q\}$

$\equiv \sim [\sim (p \rightarrow q) \vee q]$ $\{\because (p \rightarrow q) \equiv \sim p \vee q\}$

$\equiv \sim [(p \wedge \sim q) \vee q]$ $\{\because \sim (p \rightarrow q) \equiv p \wedge \sim q\}$

$\equiv \sim [q \vee (p \wedge \sim q)]$ Commutative Law

$\equiv \sim [(q \vee p) \wedge (q \vee \sim q)]$ Distributive Law

$\equiv \sim [(q \vee p) \wedge T_0]$ $\{\because q \vee \sim q$ is a Tautology $\}$

$\equiv \sim (q \vee p)$ Identity Law

Problem 10 Show that $[(p \vee q) \wedge \sim \{\sim p \wedge (\sim q \vee \sim r)\}] \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$ is a Tautology.

Solution : Let $w = u \vee v$

where $u = (p \vee q) \wedge \sim \{\sim p \wedge (\sim q \vee \sim r)\}$ and

$v = (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$

Now

$u \equiv (p \vee q) \wedge \sim \{\sim p \wedge (\sim q \vee \sim r)\}$

$\equiv (p \vee q) \wedge \sim \{\sim p \wedge \sim (q \wedge r)\}$

$\equiv (p \vee q) \wedge \sim \{\sim (p \vee (q \wedge r))\}$

$\equiv (p \vee q) \wedge \{p \vee (q \wedge r)\}$

$\equiv p \vee \{q \wedge (q \wedge r)\}$

$\equiv p \vee (q \wedge r)$

Now

$v \equiv (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$

$\equiv \sim (p \vee q) \vee \sim (p \vee r)$

$\equiv \sim \{(p \vee q) \wedge (p \vee r)\}$

$\equiv \sim \{p \vee (q \wedge r)\} \equiv \sim u$

$\therefore \quad w \equiv u \vee v \equiv u \vee \sim u$, which is always true.

$\therefore$ The given compound proposition is a Tautology.

6 Rules of Inference

To construct proofs, we need a means of drawing conclusions or deriving new assertions from old ones; this is done using rules of inference. Rules of inference specify which conclusions may be inferred legitimately from assertions known, assumed, or previously established.

1. ***Modus Ponens or Rule of detachment*** : If the statement p is assumed as true, and also the statement $p \to q$ is accepted as true, then in these circumstances, we must accept q as true. Symbolically we write it as

$$\begin{array}{l} p \\ \underline{p \to q} \\ \therefore \quad q \end{array}$$

   or $p \wedge (p \to q) \to q$ is a tautology.

   Example : It is 11.00 o'clock at Hyderabad.
   If it is 11.00 o'clock at Hyderabad, then it is 9.30 o'clock at Newyork.
   Then by rule of detachment, we can conclude that
   It is 9.30 o'clock at Newyork.

2. ***Law of hypothetical syllogism or Transitive rule*** : Whenever the two implications $p \to q$ and $q \to r$ are accepted as true, we must accept the implication $p \to r$ as true. Symbolically we write it as

$$\begin{array}{l} p \to q \\ \underline{q \to r} \\ \therefore \quad p \to r \end{array}$$

   This is because of the implication $(p \to q) \wedge (q \to r) \to (p \to r)$ is a tautology.
   This can be extended to a larger number of variables, for example

$$\begin{array}{l} p \to q \\ \underline{q \to r} \\ r \to s \\ \therefore \quad p \to s \end{array}$$

3. ***Law of contrapositive*** : Symbolically it can be written as

$$(p \to q) \Longleftrightarrow \sim q \to \sim p$$

4. ***Demorgan Laws*** : Symbolically these laws can be written as

$$\sim (p \vee q) \Longleftrightarrow \sim p \wedge \sim q$$

$$\sim (p \wedge q) \Longleftrightarrow \sim p \vee \sim q$$

7 Methods of Proof of an Implication

The propositions that commonly appear in Mathematical discussions are conditionals of the form $p \to q$, where p and q are simple or compound propositions which may involve quantifiers. Given such a conditional, the process of establishing that the conditional is true by using the laws of logic (rules) and other known facts is called a *proof of conditional*. The process of establishing that a proposition is a false is called *disproof*.

1. ***Trivial Proof*** : If it is possible to establish that q is true, then regardless of the truth values of p, $p \rightarrow q$ is true.
2. ***Vacuous Proof*** : If p is shown to be false, then $p \rightarrow q$ is true for any proposition q.
3. ***Direct Proof*** : The direct proof begins by assuming p is true and then, from available information, from the frame of reference, the conclusion q is shown to be true by valid inference.
4. ***Indirect Proof*** : The implication $p \rightarrow q$ is equivalent to the implication $\sim q \rightarrow \sim p$. Hence we can establish the truth of $p \rightarrow q$ by prooving $\sim q \rightarrow \sim p$, which is called indirect proof. Hence, in the indirect proof
   (a) assume q is false
   (b) prove on the basis of available information, p is false.
5. ***Proof by Contradiction*** : We know the fact that $p \rightarrow q$ is true if and only if $p \wedge \sim q$ is false. This is derived from DeMorgan laws and the fact $p \rightarrow q \equiv \sim p \vee q$. Hence, in the proof by contradiction
   (a) assume $p \wedge \sim q$ is true
   (b) Discover some conclusion that is false or violates the existing / established rules.
   (c) Then based on the contradiction obtained in (b), we can say $p \wedge \sim q$ is false and hence $p \rightarrow q$ is true.
6. ***Proof by Cases*** : If Proposition p is in the form $p_1 \vee p_2 \vee p_3 \vee \ldots\ldots \vee p_n$ then $(p_1 \vee p_2 \vee p_3 \vee \ldots\ldots \vee p_n) \rightarrow q$ can be established by proving separately $p_1 \rightarrow q$, $p_2 \rightarrow q$, $p_3 \rightarrow q$, ... $p_n \rightarrow q$.
7. ***Proof by Elimination of Cases*** : This method of proof is nothing more than the law of disjunctive syllogism, given by $[(p \vee q) \wedge \sim p] \rightarrow q$. This can be extended to any finite number of cases.
   $\{[(p_1 \vee p_2 \vee p_3 \vee p_4 \ldots\vee p_n) \vee q] \wedge \sim p_1 \wedge \sim p_2 \wedge \sim p_3 \wedge \sim p_4 \ldots \wedge \sim p_n\} \rightarrow q$
8. ***Conditional Proof*** : We know that $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$. Hence the proof of conditional $p \rightarrow (q \rightarrow r)$ is as follows.
   (a) combine the two antecedents p and q
   (b) Then prove r on the basis of available information.
9. ***Proof by Equivalence*** : To prove $p \leftrightarrow q$, we break it into two parts i.e., $p \rightarrow q$ and $q \rightarrow p$ and prove them separately. To prove these parts we can choose any one of the previous methods.

8 First Order Logic and Other Methods of Proof

Everyone enrolled in MRUH has lived in a hostel.
Sahithya has never lived in a hostel.
——————————————————

∴ Sahithya has never enrolled in MRUH.

Our target is to determine whether the aforementioned argument is correct or not. All three are statements and are independent as no implication is there. Propositional logic is not enough to handle these kinds of arguments therefore we'll move to first-order logic. We'll identify the validity of such arguments using first-order logic or predicate logic. So we'll understand **Predicates** and **Quantifiers**.

Predicates: Predicates are statements involving variables which are neither True nor False untill or unless the values of variables are specified.
Example:

1. x is an animal.
2. x is greater than 4.
3. x is less than 5
4. $x + y = 7$.

All the above statements are neither True nor False. They are not propositions because we can't identify thruth values for the argument.
In Predicate logic, a statement is devided into two parts: subject and predicate. Usually we denote such statements with shorthand notation; $G(x)$.

$$G(x) : x \text{ is greater than } 3,$$

where $G()$ denotes the predicate "is greater than 3" and x denotes the subject or variable. After assigning the values of a variable x, the statement $G(x)$ becomes proposition and has a truth value (either True or False)

$$G(5) : 5 \text{ is greater than } 3, \quad \text{(True)}$$

$$G(3) : 3 \text{ is greater than } 3, \quad \text{(False)}$$

Quantifiers: Quantifiers are words that refer to quantities such as "some" or "all". It tells for howmany elements a given Predicate is True. **In English**, Quantifiers are used to express the quantities without giving an exact number.
Ex. all. some, many, none, few etc.
Sentence like: —

1. "Jack has many friends"
2. "Can I have some water?"

Types of quantifiers are:

1. **Universal quantifier:** The Universal Quantifiers of $P(x)$ is the statement $P(x)$ *is true for all values of x in the domain*.

Example: Let $P(x)$ be a statement $x + 1 > x$

$$P(1) = 1 + 1 > 1 \quad = 2 > 1 \quad \text{True}$$

$$P(2) = 2 + 1 > 1 \quad = 3 > 1 \quad \text{True}$$

$P(x)$ is "True" for all natural numbers. Mathematically, we write: $(\forall x \in \mathrm{N}), P(x)$

Symbolically, we can write this as $\forall x, P(x)$, here "$\forall$" is a universal quantifier.

2. **Existential quantifier** The Existential Quantifiers of $Q(x)$ is the statement $Q(x)P$ *is True for some values of x in the domain*.

Example: Let $Q(x)$ be a statement $x < 2$

$$Q(1) = 1 < 2 \quad \text{True}$$

$$Q(2) = 2 < 2 \quad \text{False}$$

$$Q(3) = 3 < 2 \quad \text{False}$$

There exists some$(x = 1)$ in natural numbers for which $Q(x)$ is "True".

Symbolically, we can write this as $\exists x,\ Q(x)$, here "$\exists$" is an existential quantifier.

Remark: If we use a quantifier that appears within the scope of another quantifier, then it is called *nested quantifier*.

Example :

1. $O(x)$: x is an odd number; $P(x)$: x is a Prime number.
2. $B(x)$: x is a Bird.
3. $F(x)$: x can Fly.

Then "*all birds can fly*" can be written as

$$\forall x, B(x) \rightarrow F(x)$$

and "*not all birds can fly*" can be written as

$$\sim [\ \forall x, B(x) \rightarrow F(x)\ ] \text{ or } \exists x, B(x) \wedge \sim F(x)$$

4. There is a student who likes Mathematics but not Physics

$M(x)$: x likes Mathematics

$P(x)$: x likes Physics

$S(x)$: x is a student,

then the above statement can be written as $\exists x, [\ S(x) \wedge M(x) \wedge \sim P(x)\ ]$

The following table gives us the abbreviated meaning of the quantified statements.

No.	Sentence	Abbreviated Meaning
1	$\forall x,\ F(x)$	all True
2	$\exists x,\ F(x)$	at least one true
3	$\sim [\ \exists x,\ F(x)\ ]$	none true
4	$\forall x,\ [\sim F(x)\ ]$	all false
5	$\exists x,\ [\sim F(x)\ ]$	at least one false
6	$\sim [\ \exists x,\ [\sim F(x)\ ]\ ]$	none false
7	$\sim [\ \forall x,\ F(x)\ ]$	not all true
8	$\sim [\ \forall x,\ [\sim F(x)\ ]\ ]$	not all false

These can be further reduced to four, by listing the following equivalences.

No.	Sentence	Reference Numbers
1	$\forall x,\ F(x) \quad \equiv \quad \sim [\ \exists x,\ [\sim F(x)\ ]\ ]$	1 & 6
2	$\exists x,\ F(x) \quad \equiv \quad \sim [\ \forall x,\ [\sim F(x)\ ]\ ]$	2 & 8
3	$\sim [\ \exists x,\ F(x)\ ] \quad \equiv \quad \forall x,\ [\sim F(x)\ ]$	3 & 4
4	$\exists x,\ [\sim F(x)\ ] \quad \equiv \quad \sim [\ \forall x,\ F(x)\ ]$	5 & 7

Remark: The variable present in a quantified statement is called *bound variable*.

Other Methods of Proof: We can write a few more proof techniques as:

1. **Proof by example:** To show $x\exists,\ F(x)$ is true, it is sufficient to show $F(c)$ is true for some c in the universe. This type is the only situation where an example proves anything.

2. **Proof by exhaustion:** A statement of the form $\forall x,\ [\sim F(x)]$ that $F(x)$ is false for all x (all false) or, equivalently, that there are no values of x for which $F(x)$ is true (none true) will have been proven after all the objects in the universe have been examined and none found with property $F(x)$.

3. **Proof by counterexample:** To show that $\forall x,\ F(x)$ is false, it is sufficient to exhibit a specific example c in the universe such that $F(c)$ is false. This one value c is called a counterexample to the assertion $\forall x,\ F(x)$. In actual fact, a counterexample disproves the statement $\forall x,\ F(x)$.

9 Rules of Inference (Contd..)

5 ***Universal Specification*** : If a statement of the form $\forall x,\ P(x)$ is assumed to be true, then the Universal quantifier can be dropped to obtain $P(c)$ is true for an arbitrary object c in the Universe. This can be represented as

$$\frac{\forall x,\ P(x)}{\therefore \quad P(c) \text{ for all } c}$$

6 ***Universal Generalization*** : If a statement $P(c)$ is true for each element c of the Universe, then the Universal quantifier may be prefixed to obtain $\forall x, P(x)$. This can be represented as

$$\frac{P(c) \text{ for all } c}{\therefore \quad \forall x, P(x)}$$

7 ***Existential Specification*** : If $\exists x, P(x)$ is assumed to be true, then there is an element c in the universe such that $P(c)$ is true. This can be represented as

$$\frac{\exists x, P(x)}{\therefore \quad P(c) \text{ for some } c}$$

8 ***Existential Generalization*** : If $P(c)$ is true for some element c in the Universe, then $\exists X, p(X)$ is true. This can be represented as

$$\frac{P(c) \text{ for some } c}{\therefore \quad \exists x, P(x)}$$

Example : 1. All Men are fallible
All Kings are Men
Therefore, all Kings are fallible

Let, $M(x)$: x is a Men
$F(x)$: x is fallible
$K(x)$: x is a King

The above argument is symbolized as

$$\forall x, M(x) \to F(x)$$
$$\underline{\forall x, K(x) \to M(x)}$$
$$\therefore \forall x, K(x) \to F(x)$$

The proof of this can be written as follows.

	Assertion	Reason
1.	$\forall x, M(x) \to F(x)$	Premise 1
2.	$M(c) \to F(c)$	Step 1 and Rule 5
3.	$\forall x, K(x) \to M(x)$	Premise 2
4.	$K(c) \to M(c)$	Step 3 and Rule 5
5.	$K(c) \to F(c)$	Step 2, 4 and Rule 2
6.	$\forall x, K(x) \to F(x)$	Step 5 and Rule 6

Example : 2. Lions are dangerous animals
There are Lions
Therefore, there are dangerous animals

Let, $L(x)$: x is a Lion
$D(x)$: x is dangerous

$\therefore$ The above argument is symbolized as

$$\forall x, L(x) \rightarrow D(x)$$
$$\underline{\exists x, L(x) \qquad\qquad}$$
$$\therefore \exists x, D(x)$$

The proof of this can be written as follows.

	Assertion	Reason
1.	$\exists x, L(x)$	Premise 2
2.	$L(c)$	Step 1 and Rule 7
3.	$\forall x, L(x) \rightarrow D(x)$	Premise 1
4.	$L(c) \rightarrow D(c)$	Step 3 and Rule 5
5.	$D(c)$	Step 2, 4 and Rule 1
6.	$\exists x, D(x)$	Step 5 and Rule 8

10 Proof by Mathematical Induction

Let $P(n)$ be a statemment which may be either true or false for each integer n. To prove $P(n)$ is true for all integers $n \geq 1$, the following conditions are required:

(i) $P(1)$ is true.

(ii) For all $k \geq 1$, $P(k)$ implies $P(k+1)$.

If we generalise this process, then there are 3 steps to a proof using the principle of mathematical induction:

1. **Basis of Induction :** Show $P(n_0)$ is true where n_0 is starting point.
2. **Inductive hypothesis :** Assume $P(k)$ is true for $k \geq n_0$.
3. **Inductive step :** Show that $P(k+1)$ is true on the basis of the inductive hypothesis.

Example: Prove the formula $\frac{n(n+1)}{2}$ for the sum of the first n positive integer.

Proof: Let $S(n) = 1 + 2 + 3 + ... + n$. The proof consists of following steps:

1. **Basis of Induction :** Since $S(1) = 1 = 1(1 + 1)/2$, the formula is true for $n = 1$.
2. **Inductive hypothesis :** Assume the statement $S(n)$ is true for $n = k$, that is, $S(k) = 1 + 2 + 3 + ... + k = \frac{k(k+1)}{2}$.
3. **Inductive step :** Now, show that the formula is true for $n = k + 1$, that is, $S(k + 1) = \frac{(k+1)(k+2)}{2}$ follows from the inductive hypothesis.

   To do this, we have $S(k + 1) = 1 + 2 + 3 + ... + k + (k + 1) = S(k) + (k + 1)$

   Now, using inductive hypothesis $S(k) = \frac{k(k+1)}{2}$ in above, we can write

   $$S(k+1) = S(k) + (k+1) = \frac{k(k+1)}{2} + (k+1) = (k+1)(\frac{k}{2} + 1)$$

$$S(k+1) = \frac{(k+1)(k+2)}{2}$$

Hence the formula holds for $(k + 1)$ and the proof is complete by the principle of mathematical induction.

Strong Mathematical Induction: Let $P(n)$ be a statement which may be either true or false for each integer n. Then $P(n)$ is true for all positive integers if there is an integer $q \geq 1$ such that

1. $P(1), P(2),\ldots,P(q)$ are all true.
2. When $k \geq q$, the assumption that $P(i)$ is true for all integers $1 \leq i \leq k$ implies that $P(k+1)$ is true.

As in the case of the principle of mathematical induction, this form can be modified to apply to statements in which the starting value is an integer different from 1.

Thus, just as before, there are 3 steps to proof by strong mathematical induction:

1. **Basis of Induction :** Show $P(1), P(2),\ldots,P(q)$ are all true.
2. **Strong Inductive hypothesis :** Assume $P(i)$ is true for all integers i such that $1 \leq i \leq k$ where $k \geq q$.
3. **Inductive step :** Show that $P(k+1)$ is true on the basis of the strong inductive hypothesis.

Remark: In a proof using the principle of mathematical induction we are allowed to assume $P(k)$ in order to establish $P(k + 1)$. But in using strong mathematical induction we assume not only $P(k)$ but also $P(k-1), P(k-2),\ldots,P(1)$ as well, to establish $P(k+1)$.

Example: Prove that the function $b(n) = 2(3)^n - 5$ is the unique function defined by

(1) $b(0) = -3$**,** $b(1) = 1$**, and**

(2) $b(n) = 4b(n-1) - 3b(n-2)$ **for** $n \geq 2$**.**

Solution: First it is easy to check that $b(n) = 2(3)^n - 5$ satisfies the given relations.

Next, we claim that if $a(n)$ is any other function satisfying relations (1) and (2), then $a(n) = b(n)$ for all n.

Let $P(n)$ be the statement: $a(n) = b(n)$.

We prove $P(n)$ is a true statement for all non-negative integers n by strong induction.

1. **Basis of Induction :** Since we know $b(0) = 3-$and $b(1) = 1$ and we are assuming $a(0) = -3$ and $a(1) = 1$. Thus, $a(0) = b(0)$ and $a(1) = b(1)$. Therefore, $P(0)$ and $P(1)$ are true.

2. **Strong Inductive hypothesis :** Assume $P(i)$ is true for all integers i such that $0 \leq i \leq k$ where $k \geq 1$. In other words, assume $a(i) = b(i)$ for all integers $0 \leq i \leq k$ where $k \geq 1$.

3. **Inductive step :** Now, show that $P(k+1)$ is true or $a(k+1) = b(k+1)$.

From (2), we can write $a(k) = 4a(k-1) - 3a(k-2)$

or, $a(k+1) = 4a(k) - 3a(k-1)$

But, by strong inductive hypothesis,

$a(k) = b(k)$ and $a(k-1) = b(k-1)$ since $k \geq 1$ and $k - 1 \geq 0$.

Thus, $a(k+1) = 4(2(3^k) - 5) - 3(2(3^{k-1}) - 5)$

$= (8)(3^k) - (6)(3^{k-1}) - 5$

$= (8)(3^k) - 2(3^k) - 5$

$= (6)(3^k) - 5$

$= (2.3)(3^k) - 5$

$= (2)(3^{k+1}) - 5$

$= b(k+1)$

and the result is proved by induction.

If you have any queries, comments or suggestions for the improvement of this note, please write to me at *abhinava@mallareddyuniversity.ac.in*. I shall be happy to update it.

Faculty and Course Coordinator – Dr. Abhinava Srivastav

UNIT-III: Elementary Combinatorics

Syllabus

Basics of Counting, Combinations and Permutations, Enumeration of Combinations and Permutations, Enumerating Combinations and Permutations with Repetitions, Enumerating Permutations with Constrained Repetitions, Binomial Coefficients, The Binomial and Multinomial Theorems

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1 Introduction

For most applications of computers to problems,one normally needs to know, at least approximately, how much storage will be required and about how many operations are necessary. A major component of estimating the storage needed may be determining the number of items of a particular type that have to be stored. Similarly, a knowledge of how many operations the computation involves will help in assessing the length of program execution time, and thereby aid in determining the potential cost of the computation. Being able to answer such questions of the form "*How many*?", is important if one attempts to compare different methods of computation or even to decide whether or not a given computation is feasible.

Combinatorics is concerned with:

- Arrangements of elements in a set into patterns satisfying specifc rules, generally referred to as discrete structures. Here *discrete* (as opposed to continuous) typically also means finite, although we will consider some infnite structures as well.
- The existence, enumeration, analysis and optimization of discrete structures.
- Interconnections, generalizations- and specialization-relations between several discrete structures.

2 Basics of Counting

If X is a set, let us use $|X|$ to denote the number of elements in X.

2.1 Two Basic Counting Principles

The following two elementary principles act as "building blocks" for all counting problems:

2.1.1 Sum Rule: The principle of disjunctive counting

If a set X is the union of disjoint nonempty subsets $S_1, S_2, \ldots, S_n$ then

$$|X| = |S_1| + |S_2| + \ldots + |S_n|.$$

Note: It is important to note that the subsets $S_1, S_2, \ldots, S_n$ must have no elements in common. Moreover, since $X = S_1 \cup S_2 \cup \ldots \cup S_n$, each element of X is in *exactly* one of the subsets S_i. In other words, $S_1, S_2, \ldots, S_n$ is a partition of X.

Example: In how many ways can we draw a heart or a spade from an ordinary deck of playing cards? A heart or an ace? An ace or a king? A card numbered 2 through 10? A numbered card or a king?

Solution: Since there are 13 hearts and 13 spades, we may draw a heart or a spade in 13 + 13 = 26 ways.

We may draw a heart or an ace in 13 + 3 = 16 ways since there are only 3 aces that are not hearts.

We may draw an ace or a king in 4 + 4 = 8 ways.

There are 9 cards numbered 2 through 10 in each of 4 suits, clubs, diamonds, hearts, or spades, so we may choose a numbered card in 36 ways. (Note: we are counting aces as distinct from numbered cards.)

Thus, we may choose a numbered card or a king in 36 + 4 = 40 ways.

2.1.2 Product Rule: The principle of sequential counting

If $S_1, S_2, \ldots, S_n$ are nonempty sets, then the number of elements in the Cartesian product $S_1 \times S_2 \times \ldots \times S_n$ is the product $\prod_{i=1}^{n}|S_i|$. That is,

$$|S_1 \times S_2 \times \ldots \times S_n| = \prod_{i=1}^{n}|S_i|.$$

Example: Suppose that the license plates of a certain state require 3 English letters followed by 4 digits, (a) How many different plates can be manufactured if repetition of letters and digits are allowed? (b) How many plates are possible if only the letters can be repeated? (c) How many are possible if only the digits can be repeated? (d) How many are possible if no repetitions are allowed at all?

Solution: (a) $26^3\ 1 \cdot 0^4$ since there are 26 possibilities for each of the 3 letters and 10 possibilities for each of the 4 digits.

(b) $26^3 \cdot 10 \cdot 9 \cdot 8 \cdot 7$.

(c) $26 \cdot 25 \cdot 24 \cdot 10^4$.

(d) $26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 \cdot 7$.

2.2 Indirect Counting

It is sometimes beneficial to solve some combinatorial problems by counting indirectly, that is, by counting the complement of a set.

Example: Determine, by counting indirectly, the number of nonnegative integers less than 10^9 that contain the digit 1.

Solution: First, determine the integers less than 10^9 that *do not* contain the digit 1. We are considering 1-digit, or 2-digit, ... , up to 9-digit numbers. (Of course, a representation like 000002578 is actually a 4-digit number so we can consider 9 positions to be filled with any of the digits 0, 2, 3, 4, 5, 6, 7, 8, or 9. There are 9^9 such integers that do not contain the digit 1.

Thus, there are $10^9 - 9^9 = 612{,}579{,}511$ integers less than 10^9 that do contain the digit 1.

2.3 One-to-One Correspondence

In this counting technique the problem at hand is replaced by another problem where it is observed that there are exactly the same number of objects of the first type as there are of the second type. In other words, a one-to-one correspondence is set up between the objects of the first type with those of the second. (A one-to-one correspondence between 2 sets A and B is just a one-to-one function from A onto B.) Of course, we would hope that in the new context we can see how to count the objects of the second type more readily than those of the first type.

Example: Suppose that there are 101 players entered in a single elimination tennis tournament where any player who loses a match must drop out, and every match ends in a victory for some player, that is, there are no ties. In each round of the tournament, the players remaining are matched into as many pairs as possible, but if there is an odd number of players left someone receives a bye (which means an automatic victory for this player in this round). Enough rounds are played until a single player remains who wins the tournament. How many matches must be played in total?

Solution: *Approach 1* - The 50 winners and bye will go into the second round and pair into 25 matches and a bye. After this the 25 winners and the bye will go into the third round where there will be exactly 13 matches. The fourth round will have six matches and a bye; the fifth round, 3 matches and a bye; the sixth, 2 matches; the seventh will have 1 match and the winner of the seventh round wins the entire tournament. In total, there must be $50 + 25 + 13 + 6 + 2 + 1 = 100$ matches.

Approach 2 - Observe that there is a one-to-one correspondence between the number of matches and the number of losers. Each match has one and only one loser and each loser was eliminated in one and only one match. Therefore, the total number of matches is the same as the total number of losers. At the start there are 101 players and at the end there is only one undefeated player. Consequently, there must have been 100 losers and hence 100 matches are required to determine a winner.

Remark: One of the nice features of the second approach is that the problem and its solution can be generalized. Any single elimination tournament similar to the one above that starts with n contestants will require $n - 1$ matches in order to determine a winner.

2.4 Factorials

Definition: For each positive integer we define $n! = (n - 1).(n - 2)...3.2.1$ = the product of all integers from 1 to n.

Also define $0! = 1$. Note that $1! = 1$.

Thus, $6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$

We read $n!$ as "n factorial."

3 Combinations and Permutations

Definition : Combinations - A combination of n objects taken r at a time (called a r-combination of n objects) is an **unordered selection** of r of the objects.

Definition : Permutations - A permutation of n objects taken r at a time (called a r-permutation of n objects) is an **ordered selection or arrangement** of r of the objects.

Remark: We are simply defining the terms r-combinations and r-permutations here and have not mentioned anything about the properties of the n objects. For example, these definitions say nothing about whether or not a given element may appear more than once in the list of *n* objects.

Example: Suppose that the 5 objects from which selections are to be made are: *a, a, a, b, c*.

Then, the 3-combinations of these 5 objects are: *aaa, aab, aac, abc.*

The 3-permutations are: *aaa, aab, aba, baa, aac, aca, caa, abc, acb, bac, bca, cab, cba.*

Remark: We will use expressions like $\{3\cdot a, 2\cdot b, 5\cdot c\}$ to indicate either (1) that we have 3 + 2 + 5 = 10 objects including 3*a's,* 2*b's* and 5*c's*, or (2) that we have 3 objects *a, b, c* where selections are constrained by the conditions that *a* can be selected at most three times, *b* can be selected at most twice, and *c* can be chosen up to five times. The numbers 3, 2, and 5 in this example will be called **repetition numbers**.

Note: If there is no limit on the number of times an object can be repeated in a selection (except that imposed by the size of the selection) We use the symbol ∞ as a repetition number to mean that an object can occur an infinite number of times.

Example: The 3-combinations of $\{\infty\cdot a,\ 2\cdot b,\ \infty\cdot c\}$ are

aaa, aab, aac, abb, abc, ccc, ccb, cca, cbb.

Definition : If we are considering selections where each object has ∞ as its repetition number then we designate such selections as selections with unlimited repetitions.

In particular, a selection of *r* objects in this case will be called *r-combinations with unlimited repetitions* and any ordered arrangement of these *r* objects will be a *r-permutation with unlimited repetitions*.

Example: The 3-combinationos of *a, b, c, d* with unlimited repetitions are the 3-combinations of $\{\infty\cdot a, \infty\cdot b, \infty\cdot c, \infty\cdot d\}$
There are 20 such 3-combinations, namely:

aaa, aab, aac, aad, bbb, bba, bbc, bbd, ccc, cca, ccb, ccd, ddd, dda, ddb, ddc, abc, abd, acd, bcd

Moreover, there are $4^3 = 64$ of 3-permutations with unlimited repetitions since the first position can be filled in 4 ways (with a, b, c, or d), the second position can be filled in 4 ways, and likewise for the third position.

4 Enumeration of Combinations and Permutations

In this topic, we will only list formulas for combinations and permutations without repetitions. We will wait until later to use generating functions to give general techniques for enumerating combinations where other rules govern the selections.

Let $P(n, r)$ denotes the number of *r*-permutations of *n* elements without repetitions.

4.1 Theorem: Enumerating r-permutations without repetitions

$$P(n, r) = n(n-1)(n-2)...(n-r+1) = \frac{n!}{(n-r)!}$$

Proof: Since there are n distinct objects, the first position of a r-permutation may be filled in n ways. Next, the second position can be filled in $(n-1)$ ways since no repetitions are allowed and there are $(n-1)$ objects left to choose from. The third can be filled in $(n-2)$ ways and soon until the r^{th} position is filled in $(n-r+1)$ ways.

By applying the product rule, we conclude that

$$P(n, r) = n(n-1)(n-2)...(n-r+1)$$

Using factorial definition, we can write

$$P(n, r) = \frac{n!}{(n-r)!}$$

Note: When $r = n$, the above formula becomes

$$P(n, n) = \frac{n!}{0!} = n!.$$

When explicit reference to r is not made, we assume that all the objects are to be arranged; thus when we talk about the permutations of n objects we mean the case $r = n$.

Example: In how many ways can 7 women and 3 men be arranged in a row if the 3 men must always stand next to each other?

Solution: There are 3! ways of arranging the 3 men. Since the 3 men always stand next to each other, we treat them as a single entity, which we denote by X. Then if $W_1, W_2, ..., W_7$ represents the women, we next are interested in the number of ways of arranging $X, W_1, W_2, ..., W_7$. There are 8! permutations of these 8 objects. Hence there are (3!)(8!) permutations altogether (of course, if there has to be a prescribed order of an arrangement on the 3 men then there are only 8! total permutations).

Example: In how many ways can the letters of the English alphabet be arranged so that there are exactly 5 letters between the letters a and b?

Solution: There are $P(24, 5)$ ways to arrange the 5 letters between a and b ,2 ways to place a and b, and then 20! ways to arrange any 7-letter word treated as one unit along with the remaining 19 letters. The total is $P(24, 5)(20!)(2)$.

Linear and Circular Permutations: If the objects are being arranged in a line then the permutations are called linear permutations and if we arrange those in a circle then the permutations are called circular permutations.

In circular permutations, the number of permutations decreases.

Example: In how many ways can 5 children arrange themselves in a ring?

Solution: Here, the 5 children are not assigned to particular places but are only arranged relative to oneanother. Thus, the arrangements are considered the same if the children are in the same order clockwise. Hence, the position of child C_1 is immaterial and it is only the position of the 4 other children relative to C_1 that counts. Therefore, keeping C_1 fixed in position, there are 4! arrangements of the remaining children.

Notation: Let $C(n, r)$ denotes the number of r-combinations of n distinct objects where 1 is the repetition number for each element. Thus, if S is a set with n elements, $C(n, r)$ is the number of subsets of S with exactly r elements. Or in the terminology of the previous section, $C(n, r)$ is the number of r-combinations of n elements without repetitions.

We read $C(n, r)$ as "*n choose r*" to emphasize that selections are being made. Frequently, the terms $C(n, r)$ are called binomial coefficients because of their role in the expansion of $(x + y)^n$.

4.2 Theorem: Enumerating *r*-combinations without repetitions

$$C(n, r) = \frac{P(n, r)}{r!} = \frac{n!}{r!(n-r)!}$$

Example: In how many ways can a hand of 5 cards be selected from a deck of 52 cards?

Solution: Each hand is essentially a 5-combination of 52 cards. Thus there are $C(52, 5) = \frac{52!}{5!47!}$ such hands.

Example: How many 5-card hands consist of cards from a single suit?

Solution: For each of the 4 suits, spades, hearts, diamonds, or clubs, there are $C(13, 5)$ 5-card hands. Hence, there are a total of $4C(13, 5)$ such hands.

Example: In how many ways can a committee of 5 be chosen from 9 people?

Solution: $C(9, 5)$ ways.

Example: How many committees of 5 or more can be chosen from 9 people?

Solution: $C(9, 5) + C(9, 6) + C(9, 7) + C(9, 8) + C(9, 9)$.

Example: There are 21 consonants and 5 vowels in the English alphabet. Consider only 8-letter words with 3 different vowels and 5 different consonants, how many such words can be formed?

Solution: Choose the vowels, choose the consonants, and then arrange the 8 letters. Therefore, $C(5, 3)C(21, 5)8!$ such words can be formed.

Example: There are 30 females and 35 males in the junior class while there are 25 females and 20 males in the senior class. In how many ways can a committee of 10 be chosen so that there are exactly 5 females and 3 juniors on the committee?

Solution: Let us construct the following table:

Junior, Female	Junior, Male	Senior, Female	Senior, Male	No. of Ways of Selection
0	3	5	2	C(30,0)C(35,3)C(25,5)C(20,2)
1	2	4	3	C(30,1)C(35,2)C(25,4)C(20,3)
2	1	3	4	C(30,2)C(35,1)C(25,3)C(20,4)
3	0	2	5	C(30,3)C(35,0)C(25,2)C(20,5)

Thus, the total number of ways is the sum of the terms in the last column:
$C(30, 0)C(35, 3)C(25, 5)C(20, 2) + C(30, 1)C(35, 2)C(25, 4)C(20, 3)+$
$C(30, 2)C(35, 1)C(25, 3)C(20, 4) + C(30, 3)C(35, 0)C(25, 2)C(20, 5)$.

5 Enumerating Combinations and Permutations with Repetitions

Let $U(n, r)$ denote the number of r-permutations of n objects with unlimited repetitions and let $V(n, r)$ denote the number of r-combinations of n objects with unlimited repetitions. That is, if $a_1, a_2, ..., a_n$ are the n objects, we are counting r-combinations and r-permutations of $\{\infty \cdot a_1, \infty \cdot a_2, ..., \infty \cdot a_n\}$.

Permutations

Theorem: Enumerating r-permutations with unlimited repetitions: $U(n, r) = n^r$.

Proof: Each of the r positions can be filled in n ways and so by the product rule, $U(n, r) = n^r$.

Example: There are 25 true or false questions on an examination. How many different ways can a student do the examination if he or she can also choose to leave the answer blank?

Solution: 3^{25}.

Example: The results of 50 football games (win, lose or, tie) are to be predicted. How many different forecasts can contain exactly 28 correct results?

Solution: Choose 28 correct results $C(50, 28)$ ways. Each of the remaining 22 games has 2 wrong forecasts. Thus, there are $C(50, 28) \cdot 2^{22}$ forecasts with exactly 28 correct predictions.

Combinations

Let $a_1, a_2, ..., a_n$ be the distinct objects so that selections are made from $\{\infty \cdot a_1, \infty \cdot a_2, ..., \infty \cdot a_n\}$. Any r-combination will be of the form $\{x_1 \cdot a_1, x_2 \cdot a_2, ..., x_n \cdot a_n\}$ where $x_1, x_2, ..., x_n$ are the repetition numbers, each x_i is nonnegative, and $x_1 + x_2 + ... + x_n = r$. Conversely, any sequence of nonnegative integers $x_1, x_2, ..., x_n$ where $x_1 + x_2 + ... + x_n = r$

corresponds to an r-combination $\{x_1 \cdot a_1, x_2 \cdot a_2, ..., x_n \cdot a_n\}$.

First Observation: The number of r-combinations of $\{\infty \cdot a_1, \infty \cdot a_2, ..., \infty \cdot a_n\}$ equals the number of solutions of $x_1 + x_2 + ... + x_n = r$ in nonnegative integers.

Second Observation: The number of nonnegative integral solutions of $x_1 + x_2 + ... + x_n = r$ is equal to the number of ways of placing r indistinguishable balls in n numbered boxes.

Third Observation: The numberof ways of placing r indistinguishable balls in n numbered boxes is equal to the number of binary numbers with $(n - 1)$ $1's$ and r $0's$.

If there are x_1 balls in box number 1, x_2 balls in box number 2, ..., x_n balls in box number n, then in a corresponding binary number let there be x_1 $0's$ to the left of the first 1, x_2 $0's$ between the first and second 1, x_3 $0's$ between the second and third,..., and finally x_n $0's$ to the right of the last 1.(Two consecutive $1's$ mean that there were no balls in that box.)

Example: Suppose $r = 7$ and $n = 10$ in the above, that is, we are interested in 7-combinations of $\{\infty \cdot a_1, \infty \cdot a_2, ..., \infty \cdot a_{10}\}$. To the 7-combination $a_1a_1a_1a_4a_4a_8a_8$ we associate the solution (3, 0, 0, 2, 0, 0, 0, 2, 0, 0) of $x_1 + x_2 + ... + x_{10} = 7$. Then to the solution (3, 0, 0, 2, 0, 0, 0, 2, 0, 0) we associate the distribution of 3 balls in box 1, 0 balls in boxes 2, 3, 5, 6, 7, 9, and 10, 2 balls in box 4, and 2 balls in box 8. Then to this distribution of balls associate the binary number 0001110011110011.

Fourth Observation: The number of binary numbers with $n - 1$ $1's$ and r $0's$ is $C(n - 1 + r, r)$.

We have $n-1+r$ positions and we need only choose which r positions will be occupied by a 0, and then the remaining $n - 1$ positions are filled by $1's$.

We summarize:

Theorem: Enumerating r-combinations with unlimited repetitions:

$V(n, r)$ = The number of r-combinations of n distinct objects with unlimited repetitions.
= the number of nonnegative integral solutions to $x_1 + x_2 + ... + x_n = r$.
= the number of ways of distributing r similar balls into n numbered boxes
= the number of binary numbers with $n - 1$ one's and r zeros.

$$= C(n - 1 + r, r) = C(n - 1 + r, n - 1)$$

$$= \frac{(n - 1 + r)!}{r!\,(n - 1)!}$$

Remark: The number of r-combinations of $\{\infty \cdot a_1, \infty \cdot a_2, ..., \infty \cdot a_n\}$ is the same as the number of r-combinations of $\{r \cdot a_1, r \cdot a_2, ..., r \cdot a_n\}$.

Example: The number of 4-combinations of $\{\infty \cdot a_1, \infty \cdot a_2, \infty \cdot a_3, \infty \cdot a_4, \infty \cdot a_5\}$ is $C(5 - 1 + 4, 4) = C(8, 4) = 70$.

Example: How many different outcomes are possible by tossing 10 similar coins?

Solution: This is the same as placing 10 similar balls into two boxes labeled "heads" and "tails". Therefore, possible outcomes = $C(2-1+10\ ,10) = C(11, 10) = 11$.

Example: Enumerate the number of ways of placing 20 indistinguishable balls into 5 boxes where each box is nonempty.

Solution: First, place one ball in each of the 5 boxes. Then we must count the number of ways of distributing the 15 remaining balls into 5 boxes with unlimited repetitions. Using previous theorem, we can do this in $C(5-1+15, 15) = C(19, 15)$ ways.

Another Approach: We can also model this problem as a solution-of-an-equation problem. If x_i represents the number of balls in the i^{th} box, then we are asked to enumerate the number of integral solutions to $x_1 + x_2 + x_3 + x_4 + x_5 = 20$. where each $x_i > 0$. After distributing 1 ball into each of the 5 boxes, we then are to enumerate the number of integral solutions of $y_1 + y_2 + +y_3 + y_4 + y_5 = 15$ where each $y_i \geq 0$.

6 Enumerating Permutations with Constrained Repetitions

There are, of course, intermediate cases between selections with no repetitions and selections with unlimited repetition of the objects. Suppose that we are given a particular selection of r objects where there are some repetitions. What we desire is a formula for the number of permutations on this given selection of r objects.

Theorem: Enumerating n-permutations with constrained repetitions:

Let $q_1, q_2, ..., q_t$ are nonnegative integers such that $n = q_1 + q_2 + ... + q_t$. Also, let us consider that $a_1, a_2, ..., a_t$ are t distinct objects. Let $P(n; q_1, q_2, ..., q_t)$ denotes the number of n-permutations of the n-combination $\{q_1 \cdot a_1, q_2 \cdot a_2, ..., q_t \cdot a_t\}$, then

$$P(n; q_1, q_2, ..., q_t) = \frac{n!}{q_1!q_2!...q_t!}$$

Example: Find the number of arrangements of letters in the word TALLAHASSEE.

Solution: Since this equals the number of permutations of $\{3 \cdot A, 2 \cdot E, 2 \cdot L, 2 \cdot S, 1 \cdot H, 1 \cdot T\}$. Therefore, the required number of arrangements are

$$P(11; 3, 2, 2, 2, 1, 1,) = \frac{11!}{3!2!2!2!1!1!}$$

Example: In how many ways can 23 different books be given to 5 students so that 2 of the students will have 4 books each and the other 3 will have 5 books each?

Solution: Choose the 2 students to receive 4 books each in $C(5, 2)$ ways. Then to each such choice the 23 books can be distributed in $P(23; 4, 4, 5, 5, 5) = \frac{23!}{4!4!5!5!5!}$ ways. Thus there are $C(5, 2) \cdot \frac{23!}{4!4!5!5!5!}$ total distributions.

7 Binomial Coefficients

In mathematics, the binomial coefficients are the positive integrs that occur as coefficients in the binomial theorem.

In formulas arising from the analysis of algorithms in computer science, the binomial coefficients occur over and over again, so that a facility for manipulating them is a necessity. Moreover, different approaches to problems often give rise to formulas that are different in appearance yet identities of binomial coefficients reveal that they are, in fact, the same expressions.

Some Examples of Combinatorial Identities

7.1 Representation by factorials

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

for every pair of integers n and r where $n \geq r \geq 0$.

7.2 Symmetry property

$$C(n, r) = C(n, n-r)$$

7.3 Newton's Identity

$$C(n, r)C(r, k) = C(n, k)C(n-k, r-k) \text{ for integers } n \geq r \geq k \geq 0$$

A special case of this identity is $C(n, r)r = nC(n-1, r-1)$

Another special case of this identity is $C(n, r+1)(r+1) = (n-r)C(n, r)$

Remark: It might be instructive at this point to list some formulas for r-permutations along with a combinatorial interpretation.

$$P(n, r) = nP(n-1, r-1)$$

$$P(n, r) = P(n, r-1)(n-r+1)$$

7.4 Pascal's Identity

$$C(n, r) = C(n-1, r) + C(n-1, r-1)$$

Pascal's Identity in Permutations: $P(n, r) = rP(n-1, r-1) + P(n-1, r)$

7.5 Boundary Conditions

$$C(n, 0) = 1 = C(n, n)$$

7.6 Secondary Conditions

$$C(n, 1) = n = C(n, n-1)$$

7.7 Diagonal Summation

$$C(n, 0) + C(n + 1, 1) + C(n + 2, 2) + \ldots + C(n + r, r) = C(n + r + 1, r)$$

7.8 Row Summation

$$C(n, 0) + C(n, 1) + \ldots + C(n, r) + \ldots + C(n, n) = 2^n$$

7.9 Row Square Summation

$C(n, 0)^2 + C(n, 1)^2 + \ldots + C(n, r)^2 + \ldots + C(n, n)^2 = C(2n, n)$ for each positive integers n.

A special case of this identity is

$C(m, 0)C(n, 0) + C(m, 1)C(n, 1) + \ldots + C(m, m)C(n, n) = C(m + n, n)$ for integers $m \geq n \geq 0$.

7.10 Column Summation

$$C(r, r) + C(r + 1, r) + \ldots + C(n, r) = C(n + 1, r + 1)$$

for any positive integer $n \geq r$.

8 The Binomial and Multinomial Theorems

Any sum of two unlike symbols, such as $x + y$, is called a binomial. The binomial theorem is a formula for the powers of a binomial.

In general, the sum of t unlike things, $x_1 + x_2 + \ldots + x_t$ is called a multinomial.

8.1 The Binomial Theorem

Let n be a positive integer. Then for all x and y,

$$(x+y)^n = C(n, 0)x^n+C(n, 1)x^{n-1}y+C(n, 2)x^{n-2}y^2+\ldots+C(n, r)x^{n-r}y^r+\ldots+C(n, n)y^n$$

$$(x + y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \ldots + \binom{n}{r}x^{n-r}y^r + \ldots + \binom{n}{n}y^n$$

$$(x + y)^n = \sum_{r=0}^{n} C(n, r)x^{n-r}y^r$$

Note: The binomial coefficients $C(n, r) = \binom{n}{r}$ receive their name from their appearance in the expansion of powers of a binomial.

Example: $(x + y)^8 = C(8, 0)x^8 + C(8, 1)x^7y + C(8, 2)x^6y^2 + C(8, 3)x^5y^3$

$+C(8, 4)x^4y^4 + C(8, 5)x^3y^5 + C(8, 6)x^2y^6 + C(8, 7)xy^7 + C(8, 8)y^8$

$= x^8 + 8x^7y + 28x^6y^2 + 56x^5y^3 + 70x^4y^4 + 56x^3y^5 + 28x^2y^6$

$+8xy^7 + y^8.$

Corollary: Let n be a positive integer. Then for all x

$$(x + 1)^n = \sum_{r=0}^{n} C(n, r)x^{n-r} = \sum_{r=0}^{n} C(n, n - r)x^{n-r} = \sum_{r=0}^{n} C(n, r)x^r$$

Replacing x by $-x$, we have

$$(1 - x)^n = \sum_{r=0}^{n} C(n, r)(-x)^r = \sum_{r=0}^{n} C(n, r)(-1)^r x^r$$

8.2 The Multinomial Theorem

Let n be a positive integer. Then for all $x_1, x_2, \ldots, x_t$, we have $(x_1 + x_2 + \ldots + x_t)^n = \sum P(n; q_1, q_2, \ldots, q_t)\, x_1^{q_1} x_2^{q_2}, \ldots, x_t^{q_t}$ where the summation extends over all sets of nonnegative integers $q_1, q_2, \ldots, q_t$ where $q_1 + q_2 + \ldots + q_t = n$.

There are $C(n + t - 1, n)$ terms in the expansion of $(x_1 + x_2 + \ldots + x_t)^n$.

Example: Find the coefficient of $x_1^2 x_3 x_4^3 x_5^4$ in $(x_1 + x_2 + x_3 + x_4 + x_5)^{10}$.

Solution: There are $C(10 + 5 - 1, 10) = C(14, 10) = 1001$ terms in the expansion $(x_1 + x_2 + x_3 + x_4 + x_5)^{10}$.

The required coefficient is $P(10; 2, 0, 1, 3, 4) = \dfrac{10!}{2!0!1!3!4!} = 12600.$

If you have any queries, comments or suggestions for the improvement of this note, please write to me at *abhinava@mallareddyuniversity.ac.in*. I shall be happy to update it.

Faculty and Course Coordinator – Dr. Abhinava Srivastav

UNIT-IV: Advanced Counting Techniques

Syllabus

Recurrence Relations, Solving Linear Recurrence Relations, Divide-and-Conquer Algorithms and Recurrence Relations, Generating Functions, Inclusion-Exclusion, Applications of Inclusion-Exclusion

Faculty and Course Coordinator – Dr. Abhinava Srivastav

1 Recurrence Relation

A recurrence relation is an equation that defines a sequence based on a rule that gives the next term as a function of the previous term(s).

The simplest form of a recurrence relation is the case where the next term depends only on the immediately previous term. If we denote the n^{th} term in the sequence by x_n, such a recurrence relation is of the form $x_{n+1} = f(x_n)$ for some function f . One such example is $x_{n+1} = 2 - x_n/2$.

A recurrence relation can also be of higher order, where the term x_{n+1} could depend not only on the previous term x_n but also on earlier terms such as x_{n-1}, x_{n-2} etc. A second order recurrence relation depends just on x_n and x_{n-1} and is of the form $x_{n+1} = f(x_n, x_{n-1})$ for some function f with two inputs. For example, the recurrence relation $x_{n+1} = x_n + x_{n-1}$ can generate the Fibonacci numbers.

To generate sequence based on a recurrence relation, one must start with some initial values. For a first order recursion $x_{n+1} = f(x_n)$, one just needs to start with an initial value x_0 and can generate all remaining terms using the recurrence relation. For a second order recursion $x_{n+1} = f(x_n, x_{n-1})$, one needs to begin with two values x_0 and x_1. Higher order recurrence relations require correspondingly more initial values.

Note: A recurrence relation can be viewed as determining a discrete dynamical system.

2 Solving Linear Recurrence Relations

A wide variety of recurrence relations occur in models. Some of these recurrence relations can be solved using iteration or some other ad hoc technique. However, one important class of recurrence relations can be explicitly solved in a systematic way. These are recurrence relations that express the terms of a sequence as linear combinations of previous terms.

2.1 Linear Homogeneous Recurrence Relation with Constant Coefficients

A linear homogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form $a_n = c_1a_{n-1} + c_2a_{n-2} + \;\; + c_ka_{n-k}$, where $c_1, c_2, ..., c_k$ are real numbers, and $c_k = 0$.

The recurrence relation in the definition is **linear** because the right-hand side is a sum of previous terms of the sequence each multiplied by a function of n. The recurrence relation is **homogeneous** because no terms occur that are not multiples of the a_js. The coefficients of the terms of the sequence are all **constants**, rather than functions that depend on n. The **degree** is k because an is expressed in terms of the previous k terms of the sequence.

A consequence of the second principle of mathematical induction is that a sequence satisfying the recurrence relation in the definition is uniquely determined by this recurrence relation and the k initial conditions $a_0 = C_0, a_1 = C_1, ..., a_{k-1} = C_{k-1}$.

Example: The recurrence relation $f_n = f_{n-1} + f_{n-2}$ is a linear homogeneous recurrence relation of degree two. The recurrence relation $a_n = a_{n-5}$ is a linear homogeneous recurrence relation of degree five.

Example: The recurrence relation $a_n = a_{n-1} + a^2_{n-2}$ is not linear. The recurrence relation $H_n = 2H_{n-1} + 1$ is not homogeneous. The recurrence relation $B_n = nB_{n-1}$ does not have constant coefficients.

Note: Linear homogeneous recurrence relations are studied for two reasons. First, they often occur in modeling of problems. Second, they can be systematically solved.

Example: What is the solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$?

Solution: The characteristic equation of the recurrence relation is $r^2 - r - 2 = 0$. Its roots are $r = 2$ and $r = -1$. Hence, the sequence a_n is a solution to the recurrence relation if and only if $a_n = \alpha_1 2^n + \alpha_2(-1)^n$, for some constants α_1 and α_2. From the initial conditions, it follows that $a_0 = 2 = \alpha_1 + \alpha_2$, $a_1 = 7 = \alpha_1 \cdot 2 + \alpha_2 \cdot (-1)$.

Solving these two equations shows that $\alpha_1 = 3$ and $\alpha_2 = -1$. Hence, the solution to the recurrence relation and initial conditions is the sequence a_n with $a_n = 3 \cdot 2^n - (-1)^n$.

Example: Find an explicit formula for the Fibonacci numbers.

Solution: Recall that the sequence of Fibonacci numbers satisfies the recurrence relation $f_n = f_{n-1} + f_{n-2}$ and also satisfies the initial conditions $f_0 = 0$ and $f_1 = 1$.

The roots of the characteristic equation $r^2 - r - 1 = 0$ are $r1 = (1 + \sqrt{5})/2$ and $r_2 = (1 - \sqrt{5})/2$. Therefore, the Fibonacci numbers are given by

$$f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

for some constants α_1 and α_2. The initial conditions $f_0 = 0$ and $f_1 = 1$ can be used to find these constants. We have

$$f_0 = \alpha_1 + \alpha_2 = 0,$$

$$f_1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right) = 1.$$

The solution to these simultaneous equations for α_1 and α_2 is

$$\alpha_1 = \frac{1}{\sqrt{5}}, \alpha_2 = -\frac{1}{\sqrt{5}}.$$

Consequently, the Fibonacci numbers are given by

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2}\right)^n.$$

Note: If there is one characteristic root of multiplicity two, then $a_n = nr_0^n$ is another solution of the recurrence relation when r_0 is a root of multiplicity two of the characteristic equation.

Example: What is the solution of the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial conditions $a_0 = 1$ and $a_1 = 6$?

Solution: Do it yourself. Answer - $a_n = 3^n + n3^n$.

Example: Find the solution to the recurrence relation $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with initial conditions $a_0 = 1$, $a_1 = -2$, and $a_2 = -1$.

Solution: Do it yourself. Answer - $a_n = (1 + 3n - 2n^2)(-1)^n$.

2.2 Linear Nonhomogeneous Recurrence Relation with Constant Coefficients

A recurrence relation of the form $a_n = c_1a_{n-1} + c_2a_{n-2} + + c_ka_{n-k} + F(n)$, where $c_1, c_2, ..., c_k$ are real numbers and $F(n)$ is a function not identically zero depending only on n, is called a linear nonhomogeneous recurrence relation with constant coefficients. The recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2} + + c_ka_{n-k}$, is called the associated homogeneous recurrence relation. It plays an important role in the solution of the nonhomogeneous recurrence relation.

Example: The recurrence relation $a_n = 3a_{n-1} + 2n$ is an example of a linear nonhomogeneous recurrence relation with constant coefficients.

Example: Each of the recurrence relations $a_n = a_{n-1} + 2^n$, $a_n = a_{n-1} + a_{n-2} + n^2 + n + 1$, $a_n = 3a_{n-1} + n3^n$, and $a_n = a_{n-1} + a_{n-2} + a_{n-3} + n!$ is a linear nonhomogeneous recurrence relation with constant coefficients. The associated linear homogeneous recurrence relations are $a_n = a_{n-1}$, $a_n = a_{n-1} + a_{n-2}$, $a_n = 3a_{n-1}$, and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$, respectively.

Note: The key fact about linear nonhomogeneous recurrence relations with constant coefficients is that every solution is the sum of a particular solution and a solution of the associated linear homogeneous recurrence relation.

Example: Find all solutions of the recurrence relation $a_n = 3a_{n-1} + 2n$. What is the solution with $a_1 = 3$?

Solution: The associated linear homogeneous equation is $a_n = 3a_{n-1}$. Its solutions are $a_n^{(h)} = \alpha\, 3^n$, where α is a constant.

We now find a particular solution. Because $F(n) = 2n$ is a polynomial in n of degree one, a reasonable trial solution is a linear function in n, say, $p_n = cn + d$, where c and d are constants.

To determine whether there are any solutions of this form, suppose that $p_n = cn + d$ is such a solution. Then the equation $a_n = 3a_{n-1} + 2n$ becomes $cn + d = 3(c(n-1) + d) + 2n$.

Simplifying and combining like terms gives $(2 + 2c)n + (2d - 3c) = 0$. It follows that $cn + d$ is a solution if and only if $2 + 2c = 0$ and $2d - 3c = 0$.

This shows that $cn+d$ is a solution if and only if $c = -1$ and $d = -3/2$. Consequently, $a_n^{(p)} = -n - 3/2$ is a particular solution.

Therefore, the final solutions can be written as $a_n = a_n^{(p)} + a_n^{(h)} = -n - \frac{3}{2} + \alpha \cdot 3^n$, where α is a constant.

To find the solution with $a_1 = 3$, let $n = 1$ in the formula we obtained for the general solution. We find that $3 = -1 - \frac{3}{2} + 3\alpha$, which implies that $\alpha = 11/6$. So, the solution we seek is $a_n = -n - \frac{3}{2} + \frac{11}{6} 3^n$.

Example: Find all solutions of the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$.
Solution: Do it yourself. Answer - $a_n = \alpha_1 \cdot 3^n + \alpha_2 \cdot 2^n + \frac{49}{20} 7^n$.

3 Divide-and-Conquer Algorithms and Recurrence Relations

Many recursive algorithms take a problem with a given input and divide it into one or more smaller problems. This reduction is successively applied until the solutions of the smaller problems can be found quickly. These procedures follow an important algorithmic paradigm known as divide-and-conquer, and are called divide-and-conquer algorithms, because they divide a problem into one or more instances of the same problem of smaller size and they conquer the problem by using the solutions of the smaller problems to find a solution of the original problem, perhaps with some additional work.

Divide-and-Conquer Recurrence Relations: Suppose that a recursive algorithm divides a problem of size n into subproblems, where each subproblem is of size n/b (for simplicity, assume that n is a multiple of b; in reality, the smaller problems are often of size equal to the nearest integers either less than or equal to, or greater than or equal to, n/b). Also, suppose that a total of $g(n)$ extra operations are required in the conquer step of the algorithm to combine the solutions of the subproblems into a solution of the original problem. Then, if $f(n)$ represents the number of operations required to solve the problem of size n, it follows that f satisfies the recurrence relation $f(n) = af(n/b) + g(n)$.

This is called a divide-and-conquer recurrence relation.

Example - Binary Search The binary search algorithm reduces the search for an element in a search sequence of size n to the binary search for this element in a search sequence of size $n/2$, when n is even. (Hence, the problem of size n has been reduced to one problem of size $n/2$.) Two comparisons are needed to implement this reduction (one to determine which half of the list to use and the other to determine whether any terms of the list remain). Hence, if $f(n)$ is the number of comparisons required to search for an element in a search sequence of size n, then $f(n) = f(n/2) + 2$ when n is even.

Example - Finding the Maximum and Minimum of a Sequence Consider the following algorithm for locating the maximum and minimum elements of a sequence $a_1, a_2, ..., a_n$. If $n = 1$, then a_1 is the maximum and the minimum. If $n > 1$, split the sequence into two sequences, either where both have the same number of elements or where one of the sequences has one more element than the other. The problem is reduced to finding the maximum and minimum of each of the two smaller sequences. The solution to the original problem results from the comparison of the separate maxima and minima of the two smaller sequences to obtain the overall maximum and minimum. Let $f(n)$ be the total number of comparisons needed to find the maximum and minimum elements of the sequence with n elements. We have shown that a problem of size n can be reduced into two problems of size $n/2$, when n is even, using two comparisons, one to compare the maxima of the two sequences and the other to compare the minima of the two sequences. This gives the recurrence relation $f(n) = 2f(n/2) + 2$ when n is even.

Example - Merge Sort The merge sort algorithm splits a list to be sorted with n items, where n is even, into two lists with $n/2$ elements each, and uses fewer than n comparisons to merge the two sorted lists of $n/2$ items each into one sorted list. Consequently, the number of comparisons used by the merge sort to sort a list of n elements is less than $M(n)$, where the function $M(n)$ satisfies the divide-and-conquer recurrence relation $M(n) = 2M(n/2) + n$.

4 Generating Function

Generating functions are used to represent sequences efficiently by coding the terms of a sequence as coefficients of powers of a variable x in a formal power series. Generating functions can be used to solve many types of counting problems, such as the number of ways to select or distribute objects of different kinds, subject to a variety of constraints, and the number of ways to make change for a dollar using coins of different denominations. Generating functions can be used to solve recurrence relations by translating a recurrence relation for the terms of a sequence into an equation involving a generating function. This equation can then be solved to find a closed form for the generating function. From this closed form, the coefficients of the power series for the generating function can be found, solving the original recurrence relation. Generating functions can also be used to prove combinatorial identities by taking advantage of relatively simple relationships between functions that can be translated into identities involving the terms of sequences. Generating functions are a helpful tool for studying many properties of sequences besides those described in this section, such as their use for establishing asymptotic formulae for the terms of a sequence.

Definition: The generating function for the sequence $a_0, a_1, \ldots, a_k, \ldots$ of real numbers is the infinite series $G(x) = a_0 + a_1x + + a_kx^k + \cdot\cdot\cdot = \sum_{k=0}^{\infty} a_kx^k$.

Remark: The generating function for a_k given in above definition is sometimes called the ordinary generating function of a_k to distinguish it from other types of generating functions for this sequence.

Example: The generating functions for the sequences a_k with $a_k = 3$, $a_k = k + 1$, and $a_k = 2^k$ are $\sum_{k=0}^{\infty} 3x^k$, $\sum_{k=0}^{\infty}(k + 1)x^k$, $\sum_{k=0} 2^kx^k$, respectively.

Example: What is the generating function for the sequence 1, 1, 1, 1, 1, 1?
Solution: The generating function of 1, 1, 1, 1, 1, 1 is $1 + x + x^2 + x^3 + x^4 + x^5$.

We know that $\frac{(x^6 - 1)}{(x - 1)} = 1 + x + x^2 + x^3 + x^4 + x^5$ when $x \neq 1$. Consequently, $G(x) = \frac{(x^6 - 1)}{(x - 1)}$ is the generating function of the sequence 1, 1, 1, 1, 1, 1. [Because the powers of x are only place holders for the terms of the sequence in a generating function, we do not need to worry that $G(1)$ is undefined.]

Example: The function $f(x) = \frac{1}{(1 - ax)}$ is the generating function of the sequence $1, a, a^2, a^3, \ldots$ because $\frac{1}{(1 - ax)} = 1 + ax + a^2x^2 + \cdot\cdot\cdot$ when $|ax| < 1$, or equivalently, for $|x| < 1/|a|$ for $a \neq 0$.

4.1 Counting Problems and Generating Functions

We know a few techniques to count the r-combinations from a set with n elements when repetition is allowed and additional constraints may exist. Such problems are equivalent to counting the solutions to equations of the form $e_1 + e_2 + \cdot \cdot \cdot + e_n = C$, where C is a constant and each e_i is a nonnegative integer that may be subject to a specified constraint.

Example: Find the number of solutions of $e_1 + e_2 + e_3 = 17$, where e_1, e_2, and e_3 are nonnegative integers with $2 \leq e_1 \leq 5$, $3 \leq e_2 \leq 6$, and $4 \leq e_3 \leq 7$.

Solution: The number of solutions with the indicated constraints is the coefficient of x^{17} in the expansion of
$(x^2 + x^3 + x^4 + x^5)(x^3 + x^4 + x^5 + x^6)(x^4 + x^5 + x^6 + x^7)$.

This follows because we obtain a term equal to x^{17} in the product by picking a term in the first sum x^{e_1} , a term in the second sum x^{e_2} , and a term in the third sum x^{e_3}, where the exponents e_1, e_2, and e_3 satisfy the equation $e_1 + e_2 + e_3 = 17$ and the given constraints.

It is not hard to see that the coefficient of x^{17} in this product is 3. Hence, there are three solutions.

4.2 Solution of Recurrence Relations Using Generating Functions

We can find the solution to a recurrence relation and its initial conditions by finding an explicit formula for the associated generating function.

Example: Solve the recurrence relation $a_k = 3a_{k-1}$ for $k = 1, 2, 3, \ldots$ and initial condition $a_0 = 2$.

Solution: Let $G(x)$ is generating function for the sequence a_k, *i.e.* $G(x) = \sum_{k=0}^{\infty} a_k x^k$.

First note that

$$xG(x) = \sum_{k=0}^{\infty} a_k x^{k+1} = \sum_{k=1}^{\infty} a_{k-1} x^k.$$

Using the recurrence relation, we see that

$$G(x) - 3xG(x) = \sum_{k=0}^{\infty} a_k x^k - 3\sum_{k=1}^{\infty} a_{k-1} x^k$$
$$G(x) - 3xG(x) = a_0 + \sum_{k=1}^{\infty} (a_k - 3a_{k-1}) x^k = 2,$$

because $a_0 = 2$ and $a_k = 3a_{k-1}$.

Thus, $G(x) - 3xG(x) = (1 - 3x)G(x) = 2$.

Solving for $G(x)$ shows that $G(x) = 2/(1 - 3x)$.

We know that the identity $1/(1 - ax) = \sum_{k=0}^{\infty} a^k x^k$. Therefore, we can write

$$G(x) = 2\sum_{k=0}^{\infty} 3^k x^k = \sum_{k=0}^{\infty} 2 \cdot 3^k x^k.$$

Consequently, $a_k = 2 \cdot 3^k$.

4.3 Proving Identities via Generating Functions

Combinatorial identities can be established using combinatorial proofs and generating functions.

Example: Use generating functions to show that $\sum_{k=0}^{n} C(n, k)^2 = C(2n, n)$ whenever n is a positive integer.

Solution: First note that by the binomial theorem $C(2n, n)$ is the coefficient of x^n in $(1 + x)^{2n}$.

However, we also have

$(1 + x)^{2n} = [(1 + x)^n]^2 = [C(n, 0) + C(n, 1)x + C(n, 2)x^2 + ... + C(n, n)x^n]^2.$

The coefficient of x^n in this expression is

$C(n, 0)C(n, n) + C(n, 1)C(n, n - 1) + C(n, 2)C(n, n - 2) + \;\; + C(n, n)C(n, 0).$

This equals $\sum_{k=0}^{n} C(n, k)^2$, since $C(n, n - k) = C(n, k)$. It is because both $C(2n, n)$ and $\sum_{k=0}^{n} C(n, k)^2$ represent the coefficient of x^n in $(1 + x)^{2n}$, they must be equal.

5 Inclusion-Exclusion

5.1 The Principle of Inclusion-Exclusion

The number of elements in the union of the two sets A and B is the sum of the numbers of elements in the sets minus the number of elements in their intersection.

Mathematically, $|A \cup B| = |A| + |B| - |A \cap B|.$

Example: Suppose that there are 1807 freshmen at your school. Of these, 453 are taking a course in computer science, 567 are taking a course in mathematics, and 299 are taking courses in both computer science and mathematics. How many are not taking a course either in computer science or in mathematics?

Solution: To find the number of freshmen who are not taking a course in either mathematics or computer science, subtract the number that are taking a course in either of these subjects from the total number of freshmen. Let A be the set of all freshmen taking a course in computer science, and let B be the set of all freshmen taking a course in mathematics. It follows that $|A| = 453$, $|B| = 567$, and $|A \cap B| = 299$. The number of freshmen taking a course in either computer science or mathematics is

$|A \cup B| = |A| + |B| - |A \cap B| = 453 + 567 - 299 = 721.$

Consequently, there are 1807 - 721 = 1086 freshmen who are not taking a course in computer science or mathematics.

Example: A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken at least one of Spanish, French, and Russian, how many students have taken a course in all three languages?

Solution: Do it yourself. Answer - 7.

Remark: General Form of the Principle of Inclusion-Exclusion - Let $A_1, A_2, \ldots, A_n$ be finite sets. Then

$$|A_1 \cup A_2 \cup \ldots \cup A_n| = \sum_{1 \le i \le n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j| + \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| - \ldots$$

$$+ (-1)^{n+1} |A_1 \cap A_2 \cap \ldots \cap A_n|.$$

5.2 Applications of Inclusion-Exclusion

Many counting problems can be solved using the principle of inclusion-exclusion. For instance, we can use this principle to find the number of primes less than a positive integer. Many problems can be solved by counting the number of onto functions from one finite set to another. The inclusion-exclusion principle can be used to find the number of such functions.

5.2.1 The Number of Onto Functions

The principle of inclusionexclusion can also be used to determine the number of onto functions from a set with m elements to a set with n elements.

Result: Let m and n be positive integers with $m \ge n$. Then, there are

$$n^m - C(n, 1)(n-1)^m + C(n, 2)(n-2)^m - \ldots + (-1)^{n-1} C(n, n-1) \cdot 1^m$$

onto functions from a set with m elements to a set with n elements.

Example: How many onto functions are there from a set with six elements to a set with three elements?

Solution: The required number is $3^6 - C(3, 1)2^6 + C(3, 2)1^6 = 729 - 192 + 3 = 540$.

5.2.2 Derangements

A derangement is a permutation of objects that leaves no object in its original position. The principle of inclusion-exclusion can be used to count the permutations of n objects that leave no objects in their original positions.

Example: The permutation 21453 is a derangement of 12345 because no number is left in its original position. However, 21543 is not a derangement of 12345, because this permutation leaves 4 fixed.

Result: The number of derangements of a set with n elements is

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots + (-1)^n \frac{1}{n!} \right].$$

If you have any queries, comments or suggestions for the improvement of this note, please write to me at *abhinava@mallareddyuniversity.ac.in*. I shall be happy to update it.

Faculty and Course Coordinator – Dr. Abhinava Srivastav

UNIT-V: Graphs

Syllabus

Graphs and Graph Models, Graph Terminology and Special Types of Graphs, Representing Graphs and Graph Isomorphism, Connectivity, Euler and Hamilton Paths, Shortest-Path Problems, Planar Graphs, Graph Colouring

Trees: Introduction to Trees, Applications of Trees, Tree Traversal, Spanning Trees, Minimum Spanning Trees.

Faculty & Course Coordinator - Dr. Abhinava Srivastav

1 Graphs

Definition: A graph $G = (V, E)$ consists of V , a nonempty set of vertices (or nodes) and E, a set of edges. Each edge has either one or two vertices associated with it, called its endpoints. An edge is said to connect its endpoints.

Remarks: 1. The set of vertices V of a graph G may be infinite. A graph with an infinite vertex set or an infinite number of edges is called an **infinite graph**, and in comparison, a graph with a finite vertex set and a finite edge set is called a **finite graph**. In this course we will usually consider only finite graphs.

2. A graph in which each edge connects two different vertices and where no two edges connect the same pair of vertices is called a **simple graph**.

3. Graphs that may have multiple edges connecting the same vertices are called **multigraphs**. When there are m different edges associated to the same unordered pair of vertices u, v, we also say that u, v is an edge of **multiplicity** m.

4. When edges connect a vertex to itself, then such edges are called **loops**, and sometimes we may even have more than one loop at a vertex.

5. Graphs that may include loops, and possibly multiple edges connecting the same pair of vertices or a vertex to itself, are sometimes called **pseudographs**.

Definition: A directed graph (or digraph) (V, E) consists of a nonempty set of vertices V and a set of directed edges (or arcs) E. Each directed edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to start at u and end at v.

Remarks: 1. When a directed graph has no loops and has no multiple directed edges, it is called a **simple directed graph**.

2. Directed graphs that may have multiple directed edges from a vertex to a second (possibly the same) vertex are called **directed multigraphs**. When there are m directed edges, each associated to an ordered pair of vertices (u, v), we say that (u, v) is an edge of **multiplicity** m.

Note: A graph with both directed and undirected edges is called a **mixed graph**. For example, a mixed graph might be used to model a computer network containing links that operate in both directions and other links that operate only in one direction.

Remark: This terminology for the various types of graphs is given in **Table 1**. We will sometimes use the term graph as a general term to describe graphs with directed or undirected edges (or both), with or without loops, and with or without multiple edges. At other times, when the context is clear, we will use the term graph to refer only to undirected graphs.

TABLE 1 **Graph Terminology.**

Type	*Edges*	*Multiple Edges Allowed?*	*Loops Allowed?*
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Directed and undirected	Yes	Yes

2 Graph Models

Graphs are used in a wide variety of models as following:

2.1 Social Networks

Graphs are extensively used to model social structures based on different kinds of relationships between people or groups of people. These social structures, and the graphs that represent them, are known as social networks. In these graph models, individuals or organizations are represented by vertices; relationships between individuals or organizations are represented by edges.

Example: ***Acquaintanceship and Friendship Graphs*** We can use a simple graph to represent whether two people know each other, that is, whether they are acquainted, or whether they are friends (either in the real world in the virtual world via a social networking site such as Facebook). Each person in a particular group of people is represented by a vertex. An undirected edge is used to connect two people when these people know each other, when we are concerned only with acquaintanceship, or whether they are friends. No multiple edges and usually no loops are used. (If we want to include the notion of self-knowledge, we would include loops.) The acquaintanceship graph of all people in the world has more than six billion vertices and probably more than one trillion edges!

2.2 Communication Networks

We can model different communications networks using vertices to represent devices and edges to represent the particular type of communications links of interest.

Example: ***Call Graphs*** Graphs can be used to model telephone calls made in a network, such as a long distance telephone network. In particular, a directed multigraph can be used to model calls where each telephone number is represented by a vertex and each telephone call is represented by a directed edge. The edge representing a call starts at the telephone number from which the call was made and ends at the telephone number to which the call was made.We need directed edges because the direction in which the call is made matters.We need multiple directed edges because we want to represent each call made from a particular telephone number to a second number.

2.3 Information Networks

Graphs can be used to model various networks that link particular types of information. Here, we will describe how to model the World Wide Web using a graph.

Example: ***The Web Graph*** The World Wide Web can be modeled as a directed graph where each Web page is represented by a vertex and where an edge starts at the Web page *a* and ends at the Web page *b* if there is a link on *a* pointing to *b*. Because new Web pages are created and others removed somewhere on the Web almost every second, the Web graph changes on an almost continual basis. Many people are studying the properties of the Web graph to better understand the nature of the Web.

2.4 Software Design Applications

Graph models are useful tools in the design of software.

Example: ***Module Dependency Graphs*** One of the most important tasks in designing software is how to structure a program into different parts, or modules. Understanding how the different modules of a program interact is essential not only for program design, but also for testing and maintenance of the resulting software.A module dependency graph provides a useful tool for understanding how different modules of a program interact. In a program dependency graph, each module is represented by a vertex. There is a directed edge from a module to a second module if the second module depends on the first.

2.5 Transportation Networks

We can use graphs to model many different types of transportation networks, including road, air, and rail networks, as well shipping networks.

Example: ***Airline Routes*** We can model airline networks by representing each airport by a vertex. In particular, we can model all the flights by a particular airline each day using a directed edge to represent each flight, going from the vertex representing the departure airport to the vertex representing the destination airport. The resulting graph will generally be a directed multigraph, as there may be multiple flights from one airport to some other airport during the same day.

2.6 Biological Networks

Many aspects of the biological sciences can be modeled using graphs.

Example: ***Niche Overlap Graphs in Ecology*** Graphs are used in many models involving the interaction of different species of animals. For instance, the competition between species in an ecosystem can be modeled using a niche overlap graph. Each species is represented by a vertex. An undirected edge connects two vertices if the two species represented by these vertices compete (that is, some of the food resources they use are the same). A niche overlap graph is a simple graph because no loops or multiple edges are needed in this model.

2.7 Tournaments

Graphs can also be used to model different kinds of tournaments.

Example: ***Single-Elimination Tournaments*** A tournament where each contestant is eliminated after one loss is called a single-elimination tournament. We can model such a tournament using a vertex to represent each game and a directed edge to connect a game to the next game the winner of this game played in.

3 Graph Terminology

There are some terminology that describes the vertices and edges of undirected graphs.

Definition: Two vertices u and v in an undirected graph G are called **adjacent (or neighbors)** in G if u and v are endpoints of an edge e of G. Such an edge e is called **incident** with the vertices u and v and e is said to **connect** u and v.

Definition: The set of all neighbors of a vertex v of $G = (V, E)$, denoted by $N(v)$, is called the **neighborhood** of v.

Definition: The degree of a vertex in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the **degree** of that vertex. The degree of the vertex v is denoted by $deg(v)$.

Definition: A vertex of degree zero is called **isolated**. It follows that an isolated vertex is not adjacent to any vertex. A vertex is **pendant** if and only if it has degree one. Consequently, a pendant vertex is adjacent to exactly one other vertex.

Theorem: ***The Handshaking Theorem*** Let $G = (V, E)$ be an undirected graph with m edges. Then

$$2m = \sum_{v \in V} deg(v).$$

Note: This applies even if multiple edges and loops are present.

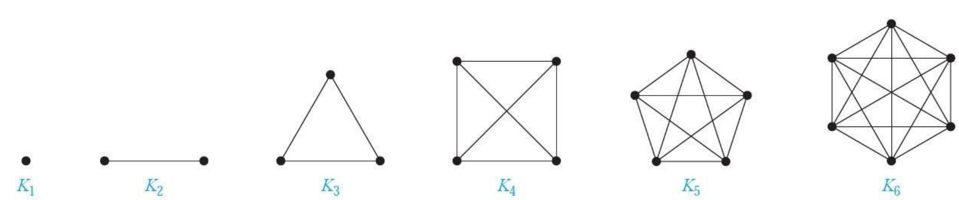
Result: An undirected graph has an even number of vertices of odd degree.

4 Special Types of Graphs

There are several classes of simple graphs. These graphs are often used as examples and arise in many applications.

4.1 Complete Graphs

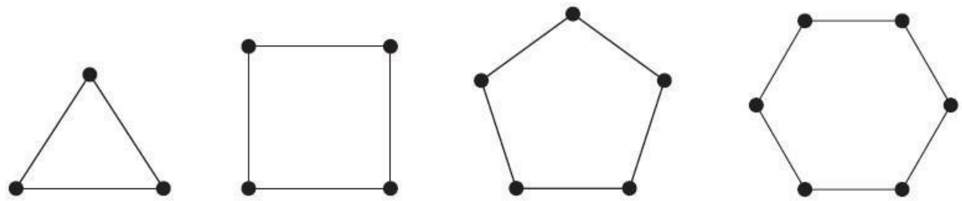
A complete graph on n vertices, denoted by K_n, is a simple graph that contains exactly one edge between each pair of distinct vertices. The graphs K_n, for $n = 1, 2, 3, 4, 5, 6$, are displayed in the following figure.



A simple graph for which there is at least one pair of distinct vertex not connected by an edge is called noncomplete.

4.2 Cycles

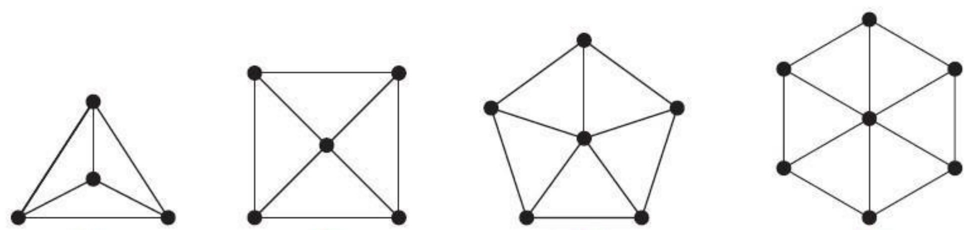
A cycle C_n, $n \geq 3$, consists of n vertices $v_1, v_2, \ldots, v_n$ and edges $v_1, v_2, v_2, v_3, \ldots, v_{n-1}, v_n,$ and v_n, v_1. The cycles C_3, C_4, C_5, and C_6 are displayed in the following figure.



C_3 C_4 C_5 C_6

4.3 Wheels

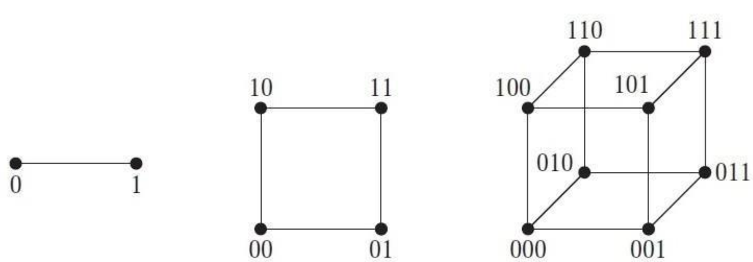
We obtain a wheel W_n when we add an additional vertex to a cycle C_n, for $n \geq 3$, and connect this new vertex to each of the n vertices in C_n, by new edges. The wheels W_3, W_4, W_5, and W_6 are displayed in the following figure.



W_3 W_4 W_5 W_6

4.4 n-Cubes

An n-dimensional hypercube, or n-cube, denoted by Q_n, is a graph that has vertices representing the 2^n bit strings of length n. Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position. We display Q_1, Q_2, and Q_3 in the following figure.

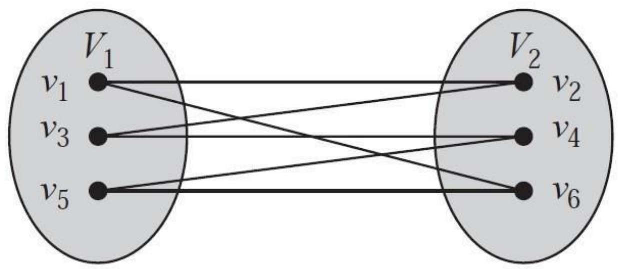


Q_1 Q_2 Q_3

4.5 Bipartite Graphs

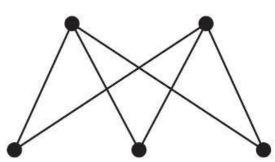
Definition: A simple graph G is called bipartite if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a bipartition of the vertex set V of G.

Example: C_6 is bipartite, as shown in the following figure, because its vertex set can be partitioned into the two sets $V_1 = v_1, v_3, v_5$ and $V_2 = v_2, v_4, v_6$, and every edge of C_6 connects a vertex in V_1 and a vertex in V_2.

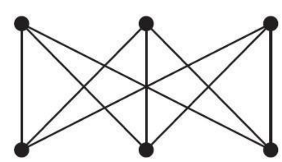


Result: A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.

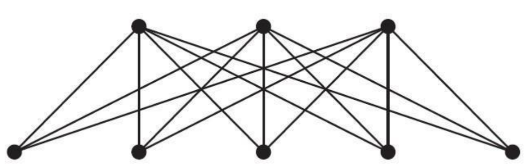
Definition: A complete bipartite graph $K_{m,n}$ is a graph that has its vertex set partitioned into two subsets of m and n vertices, respectively with an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.
The complete bipartite graphs $K_{2,3}$, $K_{3,3}$, $K_{3,5}$, and $K_{2,6}$ are displayed in the following figure.



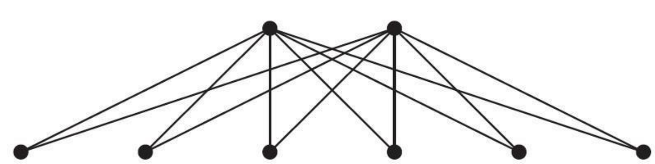
$K_{2,3}$



$K_{3,3}$



$K_{3,5}$



$K_{2,6}$

Applications of Special Types of Graphs

- Local Area Networks
- Interconnection Networks for Parallel Computation

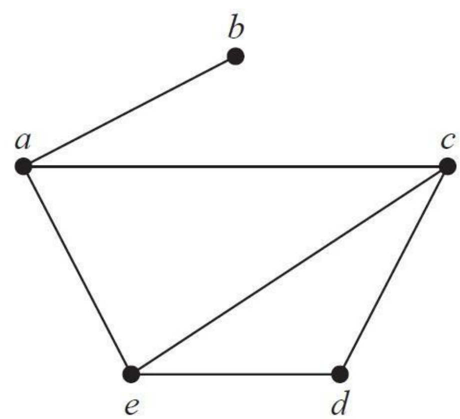
Definition: A **subgraph** of a graph $G = (V, E)$ is a graph $H = (W, F)$, where W is a subset of V and F is a subset of E.. A subgraph H of G is a proper subgraph of G if $H = G$.

Definition: Let $G = (V, E)$ be a simple graph. The **subgraph induced** by a subset W of the vertex set V is the graph (W, F), where the edge set F contains an edge in E if and only if both end points of this edge are in W.

5 Representing Graphs and Graph Isomorphism

There are many useful ways to represent graphs. One way to represent a graph without multiple edges is to list all the edges of this graph. Another way to represent a graph with no multiple edges is to use **adjacency lists**, which specify the vertices that are adjacent to each vertex of the graph.

Example: Use adjacency lists to describe the simple graph given in following figure:



Solution: Table 1 lists those vertices adjacent to each of the vertices of the graph.

TABLE 1 An Adjacency List for a Simple Graph.

Vertex	*Adjacent Vertices*
a	b, c, e
b	a
c	a, d, e
d	c, e
e	a, c, d

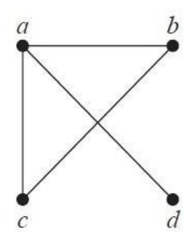
Carrying out graph algorithms using the representation of graphs by lists of edges, or by adjacency lists, can be cumbersome if there are many edges in the graph. To simplify computation, graphs can be represented using matrices. Two types of matrices commonly used to represent graphs will be presented here. One is based on the adjacency of vertices, and the other is based on incidence of vertices and edges.

5.1 Adjacency Matrices

Suppose that $G = (V, E)$ is a simple graph where $|V| = n$. Let the vertices of G are listed arbitrarily as $v_1, v_2, ..., v_n$. The adjacency matrix A (or A_G) of G, with respect to this listing of the vertices, is the nxn zero-one matrix with 1 as its $(i, j)^{th}$ entry when v_i and v_j are adjacent, and 0 as its $(i, j)^{th}$ entry when they are not adjacent. In other words, if its adjacency matrix is $A = [a_{ij}]$, then

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$

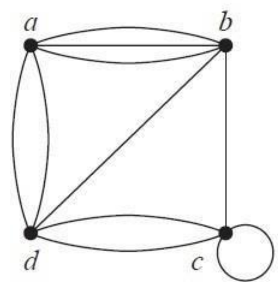
Example: Use an adjacency matrix to represent the graph shown below:



Solution: We order the vertices as *a, b, c, d*. The matrix representing this graph is

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Example: Use an adjacency matrix to represent the pseudograph shown on the next page:



Solution: The adjacency matrix using the ordering of vertices *a, b, c, d* is

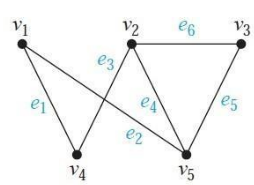
$$\begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix}$$

5.2 Incidence Matrices

Let $G = (V, E)$ be an undirected graph. Suppose that $v_1, v_2, \ldots, v_m$ are the vertices and $e_1, e_2, \ldots, e_n$ are the edges of G. Then the incidence matrix with respect to this ordering of V and E is the $m \times n$ matrix $M = [m_{ij}]_{m \times n}$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$$

Example: Represent the graph shown on the next page with an incidence matrix.



Solution: The incidence matrix is

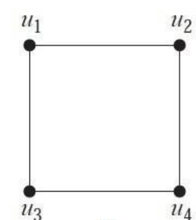
$$
\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array}
\begin{array}{c}
\begin{array}{cccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{array} \\
\begin{bmatrix}
1 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 1 \\
1 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 1 & 0
\end{bmatrix}
\end{array}.
$$

5.3 Isomorphism

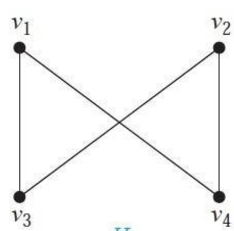
Two graphs G_1 and G_2 are *isomorphic* if there is a one-to-one correspondence between their vertex sets that preserve edges.

Other definition: Two graphs G_1 and G_2 are *isomorphic* if

- Number of vertices are same.
- Number of edges are same.
- An equal number of vertices with given degree.
- Vertex correspondence and edge correspondence is valid.



G



H

6 Connectivity

Many problems can be modeled with paths formed by traveling along the edges of graphs.

Path: A path is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along edges of the graph. As the path travels along its edges, it visits the vertices along this path, that is, the endpoints of these edges.

Circuit: A path of length $n \geq 1$ that begins and ends at the same vertex is called circuit.

Connectedness in Undirected Graphs: An undirected graph is called connected if there is a path between every pair of distinct vertices of the graph. An undirected graph that is not connected is called disconnected. We say that we disconnect a graph when we remove vertices or edges, or both, to produce a disconnected subgraph.

Connectedness in Directed Graphs: There are two notions of connectedness in directed graphs, depending on whether the directions of the edges are considered.

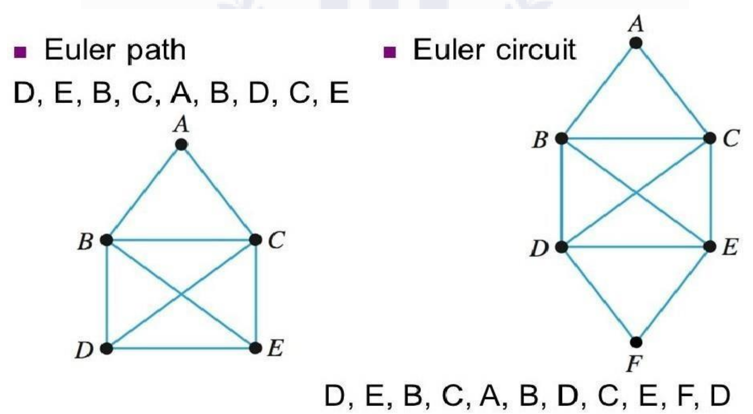
Strongly Connected: A directed graph is strongly connected if there is a path from a to b and from b to a whenever a and b are vertices in the graph.

Weakly Connected: A directed graph is weakly connected if there is a path between every two vertices in the underlying undirected graph.

7 Euler and Hamilton Paths

7.1 Euler Path and Euler Circuit

A path in a graph G is said to be **Eulerian path** if it traverses **each edge** in the graph G exactly once.
A circuit in a graph G is said to be **Eulerian circuit** if it traverses **each edge** in the graph G exactly once.

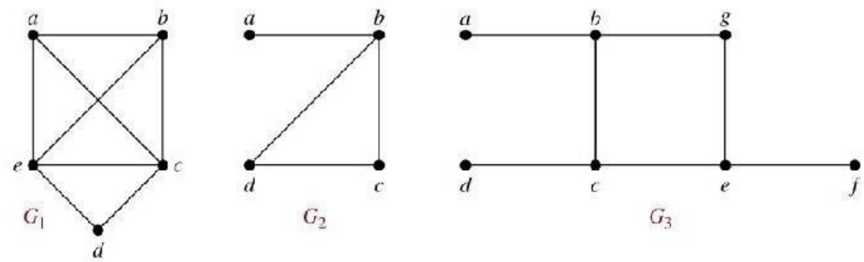


Applications of Euler Paths and Circuits: Euler paths and circuits can be used to solve many practical problems. For example, many applications ask for a path or circuit that traverses each street in a neighborhood, each road in a transportation network, each connection in a utility grid, or each link in a communications network exactly once. Finding an Euler path or circuit in the appropriate graph model can solve such problems. For example, if a postman can find an Euler path in the graph that represents the streets the postman needs to cover, this path produces a route that traverses each street of the route exactly once. If no Euler path exists, some streets will have to be traversed more than once.

7.2 Hamilton Path and Hamilton Circuit

A path which contains **every vertex** of graph *G* exactly once is called **Hamiltonian graph**.

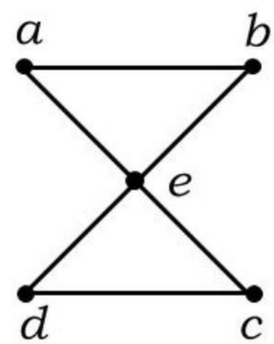
A circuit that passes through each of the **vertex** in graph *G* exactly once except starting vertex and ending vertex is called **Hamiltonian circuit**.



- G_1 has a Hamilton circuit: a, b, c, d, e, a
- G_2 does not have a Hamilton circuit, but does have a Hamilton path: a, b, c, d
- G_3 has neither.

Applications of Hamilton Paths and Circuits: Hamilton paths and circuits can be used to solve practical problems. For example, many applications ask for a path or circuit that visits each road intersection in a city, each place pipelines intersect in a utility grid, or each node in a communications network exactly once. Finding a Hamilton path or circuit in the appropriate graph model can solve such problems. The famous traveling salesperson problem or TSP asks for the shortest route a traveling salesperson should take to visit a set of cities. This problem reduces to finding a Hamilton circuit in a complete graph such that the total weight of its edges is as small as possible.

Question: Show that the following graph has Eulerian circuit but it does not have Hamiltonian circuit.

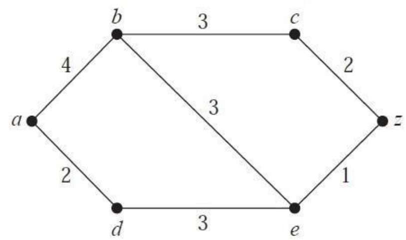


8 Shortest-Path Problems

Many problems can be modeled using graphs with weights assigned to their edges.

Weighted Graphs: Graphs that have a number assigned to each edge are called weighted graphs.

Example: What is the length of a shortest path between *a* and *z* in the weighted graph shown below?



Solution: *Dijkstra Algorithm*:

a	b	c	d	e	z	comments
0	∞	∞	∞	∞	∞	start with vertex *a* to *b* and *d*
0	4	∞	**2**	∞	∞	min{4, 2}, satrt with vertex *d* to *e*
0	**4**	∞	**2**	5	∞	min{4, 5}, satrt with vertex *b* to *c* and *e*
0	**4**	7	**2**	**5**	∞	min{7, 5}, satrting vertex *e* to *z*
0	**4**	7	**2**	**5**	**6**	reached *z*

Shortest distance between *a* to *z* is **6** and the path is {*a, d, e, z*}.

We will solve this problem by finding the length of a shortest path from a to successive vertices, until z is reached. The only paths starting at *a* that contain no vertex other than *a* are formed by adding an edge that has *a* as one endpoint. These paths have only one edge. They are *a*, *b* of length 4 and *a*, *d* of length 2. It follows that *d* is the closest vertex to *a*, and the shortest path from *a* to *d* has length 2.

We can find the second closest vertex by examining all paths that begin with the shortest path from *a* to a vertex in the set {*a, d* }, followed by an edge that has one endpoint in {*a, d* }and its other endpoint not in this set. There are two such paths to consider, *a, d, e* of length 5 and *a, b* of length 4. Hence, the second closest vertex to *a* is *b* and the shortest path from *a* to *b* has length 4.

To find the third closest vertex to *a*, we need to examine only the paths that begin with the shortest path from *a* to a vertex in the set {*a, d, b*}, followed by an edge that has one endpoint in the set {*a, d, b*} and its other endpoint not in this set. There are three such paths, *a, b, c* of length 7, *a, b, e* of length 7, and *a, d, e* of length 5. Because the shortest of these paths is *a, d, e,* the third closest vertex to *a* is *e* and the length of the shortest path from *a* to *e* is 5.

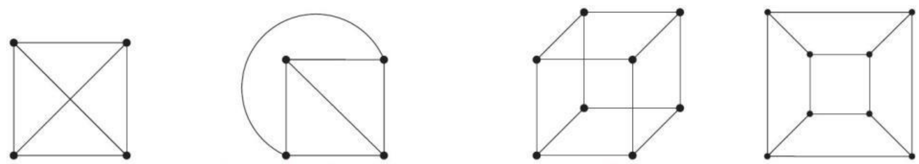
To find the fourth closest vertex to *a*, we need to examine only the paths that begin with the shortest path from *a* to a vertex in the set {*a, d, b, e*}, followed by an edge that has one endpoint in the set {*a, d, b, e*} and its other endpoint not in this set. There are two such paths, *a, b, c* of length 7 and *a, d, e, z* of length 6. Because the shorter of these paths is *a, d, e, z,* the fourth closest vertex to *a* is *z* and the length of the shortest path from *a* to *z* is 6.

9 Planar Graphs

A graph is called planar if it can be drawn in the plane without any edges crossing (where a crossing of edges is the intersection of the lines or arcs representing them at a point other than their common endpoint). Such a drawing is called a planar representation of the graph.

Note: A graph may be planar even if it is usually drawn with crossings, because it may be possible to draw it in a different way without crossings.

Example: K_4 and Q_3 are planar shown in the following figure:



K_4 K_4 with No Crossings Q_3 A Planar Representation of Q_3

Applications of Planar Graphs: Planarity of graphs plays an important role in the design of electronic circuits. We can model a circuit with a graph by representing components of the circuit by vertices and connections between them by edges.

The planarity of graphs is also useful in the design of road networks. Suppose we want to connect a group of cities by roads. We can model a road network connecting these cities using a simple graph with vertices representing the cities and edges representing the highways connecting them.

Euler's Formula: Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of regions in a planar representation of G. Then

$$r = e - v + 2.$$

Example: Suppose that a connected planar simple graph has 20 vertices, each of degree 3. Into how many regions does a representation of this planar graph split the plane?

Solution: This graph has 20 vertices, each of degree 3, so $v = 20$. Because the sum of the degrees of the vertices, $3v = 3 * 20 = 60$, is equal to twice the number of edges, $2e$, we have $2e = 60$, or $e = 30$.

Consequently, from Euler's formula, the number of regions is

$$r = e - v + 2 = 30 - 20 + 2 = 12.$$

10 Graph Colouring

A coloring of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.

Chromatic Number: The chromatic number of a graph is the least number of colors needed for a coloring of this graph. The chromatic number of a graph G is denoted by $\chi(G)$. (Here χ is the Greek letter *chi*.)

Result: The chromatic number of a planar graph is no greater than four.

Example: What is the chromatic number of K_n?

Solution: A coloring of K_n can be constructed using n colors by assigning a different color to each vertex. No two vertices can be assigned the same color, because every two vertices of this graph are adjacent. Hence, the chromatic number of K_n is n.

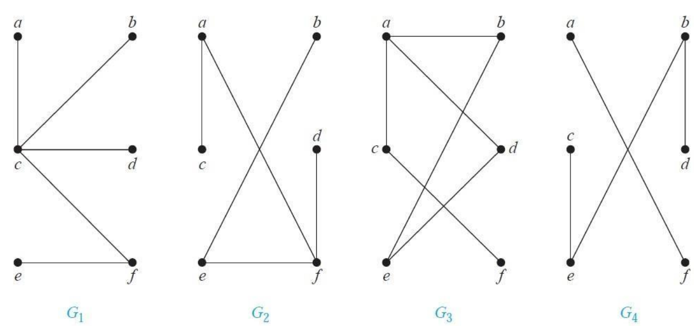
Applications of Graph Colorings : Graph coloring has a variety of applications to problems involving scheduling and assignments.

11 Trees: Introduction to Trees

Trees are particularly useful in computer science, where they are employed in a wide range of algorithms. For instance, trees are used to construct efficient algorithms for locating items in a list. They can be used in algorithms, such as Huffman coding, that construct efficient codes saving costs in data transmission and storage. Trees can be used to study games such as checkers and chess and can help determine winning strategies for playing these games. Trees can be used to model procedures carried out using a sequence of decisions. Constructing these models can help determine the computational complexity of algorithms based on a sequence of decisions, such as sorting algorithms.

Definition: A connected graph that contains no simple circuits is called a tree.

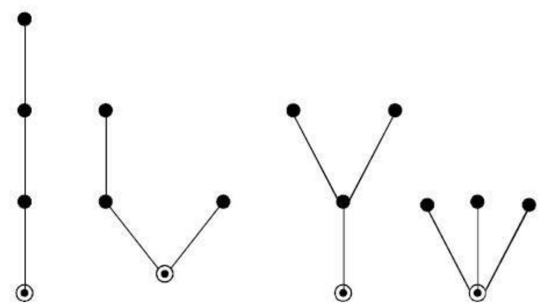
Example: Which of the graphs shown in the following figure are trees?



Solution: G_1 and G_2 are trees, because both are connected graphs with no simple circuits. G_3 is not a tree because *e, b, a, d, e* is a simple circuit in this graph. Finally, G_4 is not a tree because it is not connected.

Result: An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.

Rooted Tree: A rooted tree is a tree in which one vertex has been designated as the root and every edge is directed away from the root.

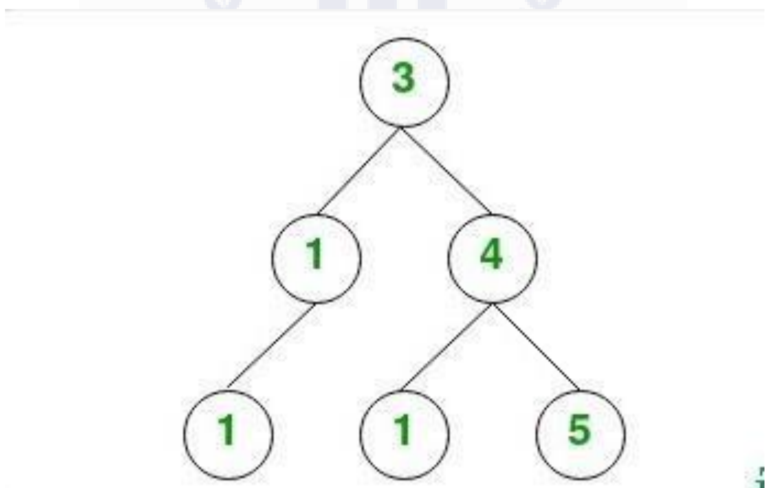


Binary Tree: A binary tree is defined as a tree in which there is exactly one vertex of degree two, and each of remaining vertices is of degree one or three. The vertex of degree two serves as a root.

Pendant Vertex in Tree: A vertex of degree one is called pendant vertex of tree. In the following graph vertices **1**, **1** and **5** are pendants.

Path Lenghth of Tree: A path length of a tree is defined as the sum of edges from the root of all pendant vertices.
Path lenght of the following tree is:

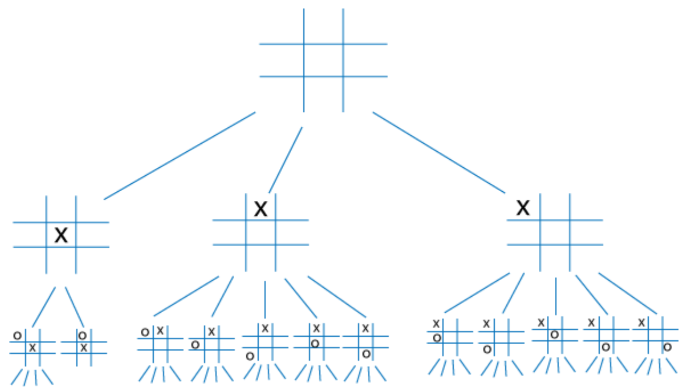
$$2 + 2 + 2 = 6$$



Result: A tree with n vertices has $n - 1$ edges.

12 Applications of Trees

- Storing hierarchical data
- Binary Search Tree
- Game Tree and many more



Example: Form a binary search tree for the words mathematics, physics, geography, zoology, meteorology, geology, psychology, and chemistry (using alphabetical order).

Solution: Figure 1 displays the steps to construct this binary search tree. The word mathematics is the key of the root. Because physics comes after mathematics (in alphabetical order), add a right child of the root with key physics. Because geography comes before mathematics, add a left child of the root with key geography. Next, add a right child of the vertex with key physics, and assign it the key zoology, because zoology comes after mathematics and after physics. Similarly, add a left child of the vertex with key physics and assign this new vertex the key meteorology. Add a right child of the vertex with key geography and assign this new vertex the key geology.Add a left child of the vertex with key zoology and assign it the key psychology. Add a left child of the vertex with key geography and assign it the key chemistry.

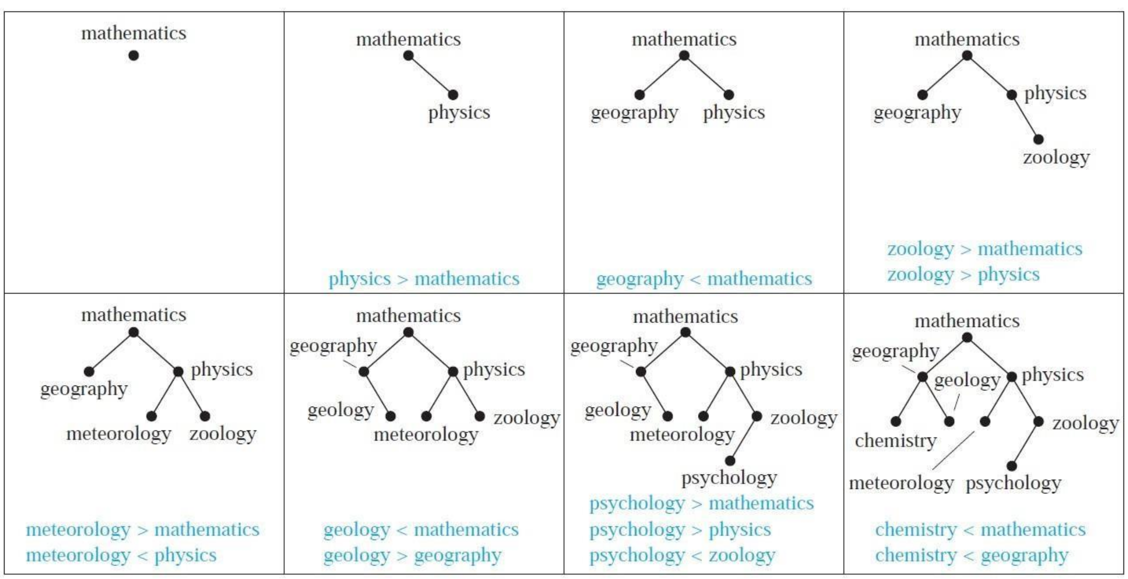


FIGURE 1 **Constructing a Binary Search Tree.**

13 Tree Traversal

Traversal Algorithms: Procedures for systematically visiting every vertex of an ordered rooted tree are called traversal algorithms.

We will describe three of the most commonly used such algorithms, preorder traversal, inorder traversal, and postorder traversal. Each of these algorithms can be defined recursively.

Preorder Traversal: Let T be an ordered rooted tree with root r. If T consists only of r, then r is the preorder traversal of T . Otherwise, suppose that $T_1, T_2, \ldots, T_n$ are the subtrees at r from left to right in T . The preorder traversal begins by visiting r. It continues by traversing T_1 in preorder, then T_2 in preorder, and so on, until T_n is traversed in preorder.

Inorder Traversal: Let T be an ordered rooted tree with root r. If T consists only of r, then r is the inorder traversal of T . Otherwise, suppose that $T_1, T_2, \ldots, T_n$ are the subtrees at r from left to right. The inorder traversal begins by traversing T_1 in inorder, then visiting r. It continues by traversing T_2 in inorder, then T_3 in inorder, . . . , and finally T_n in inorder.

Postorder Traversal: Let T be an ordered rooted tree with root r. If T consists only of r, then r is the postorder traversal of T . Otherwise, suppose that $T_1, T_2, \ldots, T_n$ are the subtrees at r from left to right. The postorder traversal begins by traversing T_1 in postorder, then T_2 in postorder, . . . , then T_n in postorder, and ends by visiting r.

Infix notation: The form of an expression (including a full set of parentheses) obtained from an inorder traversal of the binary tree representing this expression.

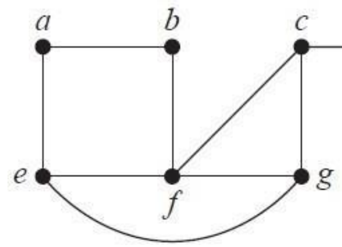
Prefix (or Polish) notation: The form of an expression obtained from a preorder traversal of the tree representing this expression.

Postfix (or reverse Polish) notation: The form of an expression obtained from a postorder traversal of the tree representing this expression.

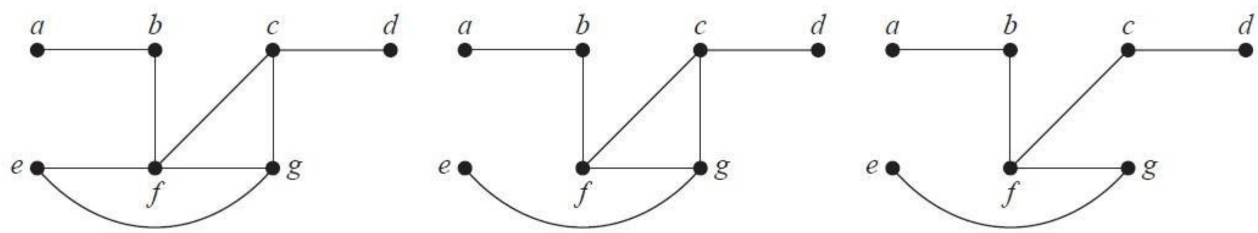
14 Spanning Trees

Definition: Let G be a simple graph. A spanning tree of G is a subgraph of G that is a tree containing every vertex of G.

Example: Find a spanning tree of the simple graph G shown below:



Solution: The graph G is connected, but it is not a tree because it contains simple circuits. Remove the edge $\{a, e\}$. This eliminates one simple circuit, and the resulting subgraph is still connected and still contains every vertex of G. Next remove the edge $\{e, f\}$ to eliminate a second simple circuit. Finally, remove edge $\{c, g\}$ to produce a simple graph with no simple circuits. This subgraph is a spanning tree, because it is a tree that contains every vertex of G. The sequence of edge removals used to produce the spanning tree is illustrated in the figure below.



Edge removed: $\{a, e\}$ $\{e, f\}$ $\{c, g\}$

(a) (b) (c)

Note: Spanning tree is not unique.

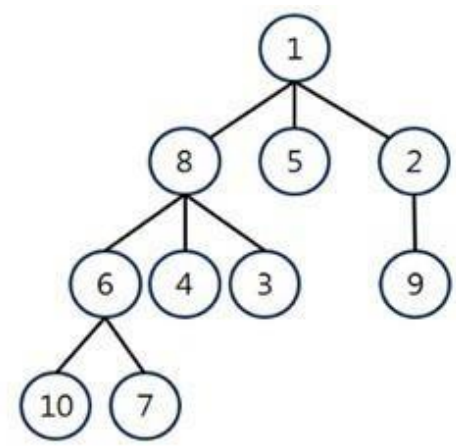
Result: A simple graph is connected if and only if it has a spanning tree.

BFS and DFS - Algorithms to produce a spanning tree

Breadth-first search: A procedure for constructing a spanning tree that successively adds all edges incident to the last set of edges added, unless a simple circuit is formed.

Working Rule:

- QUEUE – first in first out: Visit any vertex.
- Start exploring all adjacent vertices connected through the selected vertex.
- Take vertices in any order and continue exploration untill all vertices are included.

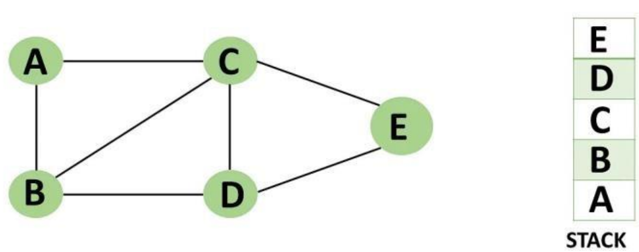


BFS: 1 8 5 2 6 4 3 9 10 7

Depth-first search, or backtracking: A procedure for constructing a spanning tree by adding edges that form a path until this is not possible, and then moving back up the path until a vertex is found where a new path can be formed.

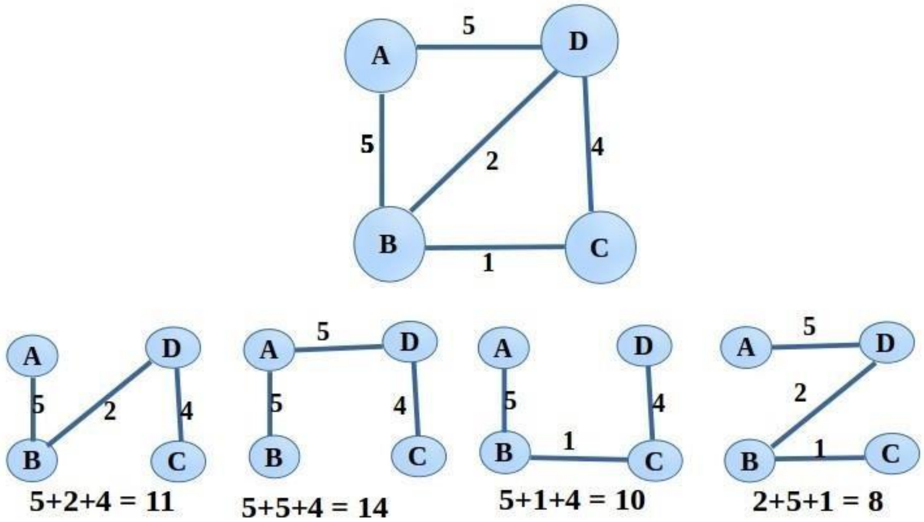
Working Rule:

- STACK – last in first out: Visit any vertex.
- Like pre-order traverse, once we go to new vertex, we will suspend this vertex and start exploring new verex .
- Continue untill all vertices are included.



15 Minimum Spanning Trees

Definition: A minimum spanning tree in a connected weighted graph is a spanning tree that has the smallest possible sum of weights of its edges.



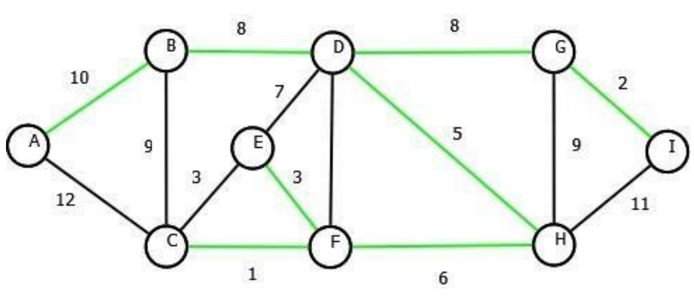
The spanning tree having weight 8 is minimal spanning tree.
Finding all spaaning tree of spanning tree is not that easy. To find minimal spanning tree we have following algorithm.

Kruskal's algorithm: A procedure for producing a minimum spanning tree in a weighted graph that successively adds edges of least weight that are not already in the tree such that no edge produces a simple circuit when it is added.

Working Rule:

- Choose an edge of minimum weight.
- At each step, choose the edge whose inclusion will not create a circuit.
- If G has n vertices, then stop after $n-1$ edges.

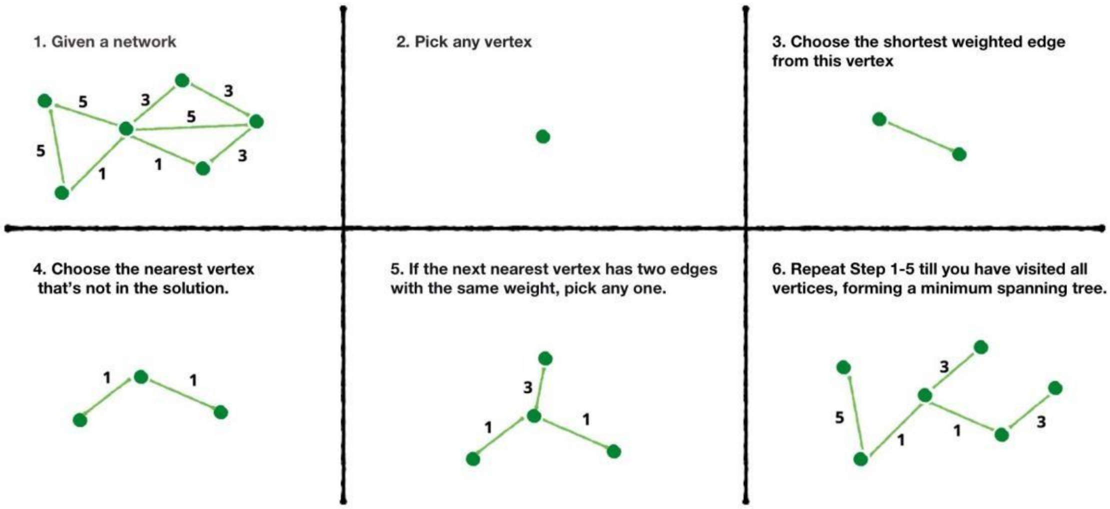
Example: In the following graph, the green colour one is minimal spanning tree (MST) by Kruskal's algorithm.



Prim's algorithm: A procedure for producing a minimum spanning tree in a weighted graph that successively adds edges with minimal weight among all edges incident to a vertex already in the tree so that no edge produces a simple circuit when it is added.

Working Rule:

- Select any vertex, choose an edge and smallest weight from *G*.
- At each stage, choose the edge of smallest weight joining a vertex already included to vertex not yet included.
- Continue untill all vertices are included.



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To the readers - If you have any queries, comments or suggestions for the further improvement of these notes, please write to *abhinava@mallareddyuniversity.ac.in*. I shall be glad to update it.

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